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RESISTORS, CURRENT SENSE RESISTORS, BATTERY SHUNTS, SHUNT RESISTORS, AND METHODS OF MAKING

Abstract

A plurality of resistors are disclosed herein. The resistor may include one or more resistive elements and a plurality of conductive portions. Openings or slots, which can be configured to adjust temperature coefficient or resistance (TCR) values of the resistor, are formed in the resistive elements. The shape, quantity, and orientation of the openings or slots can vary. In one aspect, header assemblies are provided for securing or holding pins relative to the resistors.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation of U.S. patent application Ser. No. 18/096,951, filed Jan. 13, 2023, which is a continuation of U.S. patent application Ser. No. 17/317,233, filed May 11, 2021, which claims the benefit of U.S. Provisional Patent Application 63/068,243, filed Aug. 20, 2020, the entire contents of all of which are incorporated by reference as if fully set forth herein.

FIELD OF THE INVENTION

[0002] The present invention relates to resistors, current sense resistors, battery shunts, and shunt resistors, having various arrangements of conductive portions and resistive elements.

BACKGROUND

[0003] Current sense resistors and/or shunt resistors and/or battery shunts, as they may be referred to by those of skill in the art, are known types of resistors used widely, for example, in the automotive field. These types of resistors may be used, for example and by way of illustration, to measure electric current. Their construction typically includes a flat strip of a resistive material that is coupled between high conductivity metal terminals forming the main terminals of the device.

[0004] Redundancy of systems to maintain safe operation if the primary system fails is a requirement in automotive applications. Full electric car application requirements now involve systems that use multiple traditional single element battery shunts adding to space requirements and increased cost.

[0005] Various systems utilize different methods for temperature compensation, such as added temperature sensing components and programming to compensate for the shape of the resistance vs. temperature plot which add overall expense to designs.

[0006] What is needed is an improved configuration and method of making a current sense resistor or battery shunt.

[0007] There is further a need for an improved resistor with the ability to easily, accurately, and efficiently, adjust the temperature coefficient of resistance (TCR) of the resistor.

[0008] There is further a need for the ability to sense current in battery management circuits that allows for users to connect redundant systems to the same circuit with different control or sensing units creating increased safety due to the redundancy.

SUMMARY

[0009] Various resistors and methods of manufacturing resistors are provided.

[0010] The various designs of the invention provide for an expansion and improvement of the features to improve the resistance change with temperature such as those contained in U.S. Pat. No. 8,878,643, the entire contents of which are incorporated herein by reference, and include additional ways of isolating the voltage sense points that can then be used for mounting pins inside of those features via connectors such as headers that can then be used for external connection to the sense points. This also simplifies the manufacturing process. This sense point mounting inside of the isolation feature further enhances the voltage sense isolation by concentrating the contact area inside of the thickness of the part as opposed to just on one surface.

[0011] In an aspect of the invention, the resistor comprises a first conductive portion, a first resistive element, a second conductive portion, and second resistive element, and a third conductive portion. In this aspect, the resistor may be referred to or considered a dual resistive element shunt.

[0012] The first conductive portion has a first or outer side and a second or inner side. The first resistive element has a first or outer side and a second or inner side. The second conductive portion has a first side and a second side. The second resistive element has a first or outer side and a second or inner side. The third conductive portion has a first or outer side and a second or inner side. The first conductive portion is attached at the second or inner side to the first or outer side of the first resistive element. The first conductive portion and the first resistive element may be connected, joined, bonded, or attached by various means such as welding, bonding, adhesives, or other known attachment means for bonding conductive materials and resistive materials.

[0013] The first resistive element is attached at the second or inner side to the first side of the second conductive portion. The first resistive element and the second conductive portion may be connected, joined, bonded, or attached by various means.

[0014] The second conductive portion is attached at the second side to the second or inner side of the second resistive element. The second conductive portion and the second resistive element may be connected, joined, bonded, or attached by various means.

[0015] The second resistive element is attached at the first or outer side to the second or inner side of the third conductive portion. The second resistive element and the third conductive portion may be connected, joined, bonded, or attached by various means.

[0016] In an aspect of the invention, the resistor comprises a first conductive portion, a first resistive element, and a second conductive portion. In this aspect, the resistor may be referred to or considered a redundant sense shunt, or a redundant current sense shunt.

[0017] Additional sense contact points to create a ground reference point connection for dual low or high side ground system requirements can easily be added to these headers if required by the final application.

[0018] The “openings” described herein may also be considered or referred to as “slots” or “holes.” The term “slots” may be used to refer to a type of opening that has an elongated profile in at least one direction or dimension. In some aspects, the term opening is used generically to refer to a slot or hole.

[0019] The “conductive portions” described herein may also be considered or referred to as “terminals” or “conductive terminals.”

[0020] The openings (slots or holes) in the conductive portions configured to adjust, change, impact, effect, or modify the TCR value of the resistor may be considered “TCR adjustment slots,” or “TCR compensation openings,” or “TCR slots.” These may also be referred to as “boots” or “features.”

[0021] Also shown and described are various header pins, as are known in the art. These may be referred to as “connectors,” “brackets,” or “frames,” and provide support for the conductive pins that will electrically connect the resistor to other electrical parts, components, equipment or devices.

[0022] According to various aspects of the invention, the TCR compensation openings can also be adjusted or changed in shape and size to enhance, adjust, modify, or change the temperature coefficient of resistance (TCR) in different designs and values as needed for a selected use. Such adjustments can be done in a very precise manner resulting in very low TCR values limiting the need for extra devices and programming to track the TCR plot of a given product design.

[0023] In aspects described herein, the resistors may include a plurality of resistive elements and provide for various TCR adjustment openings, slots or areas, as well as various connection points for voltage sensing, which provide what can be referred to as a redundant configuration or redundancy.

[0024] The above summaries are not intended to represent every embodiment or every aspect of the

present disclosure. Rather, the foregoing summaries merely provides an exemplification of some of the aspects and features set forth herein. The above features, as well as other features of the present disclosure, will be apparent from the following detailed description of representative embodiments and modes for carrying out the disclosure when taken in connection with the accompanying drawings and appended claims. Moreover, this disclosure expressly includes any and all combinations and subcombinations of the elements and features presented herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] FIG. 1 illustrates is a perspective view of a resistor according to an aspect of the invention.

[0026] FIG. 2 illustrates a perspective view of a resistor according to an aspect of the invention.

[0027] FIG. 3 illustrates a front view of a resistor according to an aspect of the invention.

[0028] FIG. 4 illustrates a front view of a resistor according to an aspect of the invention.

[0029] FIG. 5 illustrates a front view of a resistor according to an aspect of the invention.

[0030] FIG. 6 illustrates a front view of a resistor according to an aspect of the invention.

[0031] FIG. 7 illustrates a front view of a resistor according to an aspect of the invention.

[0032] FIG. 8 illustrates a front view of a resistor according to an aspect of the invention.

[0033] FIG. 9 illustrates a front view of a resistor according to an aspect of the invention.

[0034] FIG. 10 illustrates a flow chart according to one aspect of the invention.

[0035] FIG. 11 illustrates another flow chart according to another aspect of the invention.

[0036] FIG. 12 illustrates a flux path of a resistor according to one aspect of the invention.

[0037] FIG. 13A illustrates a front view of another embodiment of a resistor according to an aspect of the invention.

[0038] FIG. 13B illustrates a front view of another embodiment of a resistor according to an aspect of the invention.

[0039] FIG. 13C illustrates a front view of another embodiment of a resistor according to an aspect of the invention.

[0040] FIG. 13D illustrates a front view of another embodiment of a resistor according to an aspect of the invention.

[0041] FIG. 13E illustrates a front view of another embodiment of a resistor according to an aspect of the invention.

[0042] FIG. 14 is a chart illustrating measurements for a redundant shunt TCR with optimized voltage sense isolation for two elements.

[0043] FIG. 15 is a chart illustrating measurements for a shunt with different amounts of voltage sense isolation.

[0044] FIG. 16 illustrates a cylindrical resistor according to an aspect of the invention.

[0045] FIGS. 17A-17M illustrate variations of resistors according to multiple aspects of the invention.

[0046] FIG. 18 illustrates a front view of a resistor according to an aspect of the invention.

[0047] FIG. 19 illustrates a front view of a resistor according to an aspect of the invention.

[0048] FIG. 20 illustrates a front view of a resistor according to an aspect of the invention.

[0049] FIG. 21A illustrates a front view of another embodiment of a resistor according to an aspect of the invention.

[0050] FIG. 21B illustrates a cross sectional view from a side edge of the resistor of FIG. 21A.

[0051] FIG. 22 illustrates a front view of another embodiment of a resistor according to an aspect of the invention.

[0052] FIG. 23 illustrates a front view of another embodiment of a resistor according to an aspect of the invention.

[0053] FIG. **24A** illustrates a front view of another embodiment of a resistor according to an aspect of the invention.

[0054] FIG. **24B** illustrates a cross sectional view from a side edge of the resistor of FIG. **24A**.

[0055] FIG. **25** illustrates a front view of a resistor with a printed circuit board in two different states.

[0056] FIG. **26A** illustrates a front view of a resistor with a printed circuit board configured to engage pins.

[0057] FIG. **26B** illustrates a rear view of the resistor of FIG. **26A**.

[0058] FIG. **26C** illustrates a view from a side edge of the resistor of FIGS. **26A** and **26B**.

[0059] FIG. **27** illustrates a flow chart according to one aspect of the invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0060] FIG. **1** shows a resistor **10** generally formed of a first resistive element **11** disposed between a first conductive portion **12** and a second conductive portion **14**, and a second resistive element **13** disposed between the second conductive portion **14** and a third conductive portion **16**.

[0061] As oriented in the depiction of FIG. **1**, the first conductive portion **12** has a first or outer side **18** facing in a first direction **X1**, and a second or inner side **20** facing in a second direction **X2** which is a direction opposite the first direction **X1**, a third side **22** facing in a direction **Y1**, and a fourth side **24** facing in a direction **Y2** which is a direction opposite the direction **Y1**. One of ordinary skill in the art would understand that the exact orientation of the resistor **10** may vary. The first conductive portion **12** may be formed generally as a plate or a strip or a bar. The first conductive portion **12** may be formed of a conductive metal. In an embodiment, the first conductive portion **12** is formed from copper (Cu) or a Cu alloy. Other conductive materials that may be used to form the first conductive portion **12** are, either alone or in combination, tin (Sn), aluminum (Al), silver (Ag), or gold (Au), with the alloy being plated or not plated with layers of nickel (Ni), Sn, or other similar materials.

[0062] The first resistive element **11** has a first or outer side **26** facing in the first direction **X1**, and a second or inner side **28** facing in the second direction **X2**, a third side **30** facing in the direction **Y1**, and a fourth side **32** facing in the direction **Y2**. The first resistive element **11** may be formed generally as a plate or a strip or a bar. The first resistive element **11** may be formed of a resistive material, or combinations of materials including resistive materials. In an embodiment, the first resistive element **11** is formed from or comprises alloys of copper-nickel-manganese (CuNiMn), copper manganese tin (CuMnSn), copper nickel (CuNi), nickel-chromium-aluminum (NiCrAl), or nickel-chromium (NiCr), or other alloys known to those of skill in the art acceptable for use as a resistive element as part of a shunt resistor. Other resistive materials that may be used to form the first resistive element **11** are, either alone or in combination, Cu, Ni, Mn, Cr, Al, and iron (Fe).

[0063] The first conductive portion **12** is attached at the second or inner side **20** to the first or outer side **26** of the first resistive element **11**. The first conductive portion **12** and the first resistive element **11** may be attached, connected, joined, bonded, or attached by various means such as welding, bonding, adhesives, ultrasonic bonding, soldering, brazing, or other methods that provide a low resistive connection, or any other acceptable connection means providing for electrical and/or thermal communication. The first conductive portion **12** is generally located at a first side **34** or first portion of the resistor **10**.

[0064] As oriented in the depiction of FIG. **1**, the second conductive portion **14** has a first side **36** facing in the first direction **X1**, and a second side **38** facing in the second direction **X2**, a third side **40** facing in the direction **Y1**, and a fourth side **42** facing in the direction **Y2**. The second conductive portion **14** may be formed generally as a plate or a strip or a bar. The second conductive portion **14** may be formed of a conductive metal. In an embodiment, the second conductive portion **14** is formed from copper (Cu) or a Cu alloy. Other conductive materials that may be used to form the second conductive portion **14** are, either alone or in combination, Sn, Al, Ag, or Au, with the alloy being plated or not plated with layers of Ni, Sn, or other similar materials. The second

conductive portion **14** is generally located or positioned adjacent at a central location **44** or midpoint of the resistor **10**.

[0065] The second conductive portion **14** is attached at the first side **36** to the second or inner side **28** of the first resistive element **11**. Accordingly, the first conductive portion **12** and the second conductive portion **14** are oriented on opposite sides of the first resistive element **11**. The second conductive portion **14** and the first resistive element **11** may be connected, joined, bonded, or attached by various means such as welding, bonding, adhesives, ultrasonic bonding, soldering, brazing, or other methods that provide a low resistive connection, or any other acceptable connection means providing for electrical and/or thermal communication.

[0066] The second resistive element **13** has a first or outer side **46** facing in the second direction **X2**, and a second or inner side **48** facing in the first direction **X1**, a third side **50** facing in the direction **Y1**, and a fourth side **52** facing in the direction **Y2**. The second resistive element **13** may be formed generally as a plate or a strip or a bar. The second resistive element **13** may be formed of a resistive material, or combinations of materials including resistive materials. In an embodiment, the second resistive element **13** is formed from or comprises alloys of copper-nickel-manganese (CuNiMn), copper manganese tin (CuMnSn), copper nickel (CuNi), nickel-chromium-aluminum (NiCrAl), or nickel-chromium (NiCr), or other alloys known to those of skill in the art acceptable for use as a resistive element as part of a shunt resistor. Other resistive materials that may be used to form the second resistive element **13** are, either alone or in combination, Cu, Ni, Mn, Cr, Al, and Fe.

[0067] The first resistive element **11** and the second resistive element **13** may have the same resistance value, or different resistance values. It is appreciated that the resistance values the first resistive element **11** or the second resistive element **13** may be selectively adjusted based on need or function, such as by trimming, as is known in the relevant art.

[0068] The second conductive portion **14** is attached at the second or inner side **38** to the second or inner side **48** of the second resistive element **13**. The second conductive portion **14** and the second resistive element **13** may be attached, connected, joined, bonded, or attached by various means such as welding, bonding, adhesives, ultrasonic bonding, soldering, brazing, or other methods that provide a low resistive connection, or any other acceptable connection means providing for electrical and/or thermal communication.

[0069] In the arrangement as shown in FIG. **1**, the first resistive element **11** and second resistive element **13** are positioned on opposite sides of the second conductive portion **14**.

[0070] As oriented in the depiction of FIG. **1**, the third conductive portion **16** has an inner or second side **54** facing in the first direction **X1**, and a first or outer side **56** facing in the second direction **X2**, a third side **58** facing in the direction **Y1**, and a fourth side **60** facing in the direction **Y2**. The third conductive portion **16** may be formed generally as a plate or a strip or a bar. The third conductive portion **16** may be formed of a conductive metal. In an embodiment, the third conductive portion **16** is formed from copper (Cu) or a Cu alloy. Other conductive materials that may be used to form the third conductive portion **16** are, either alone or in combination, Sn, Al, Ag, or Au, with the alloy being plated or not plated with layers of Ni, Sn, or other similar materials. The second conductive portion **14** is generally located or positioned adjacent at a second side **62** or second portion of the resistor **10**.

[0071] The third conductive portion **16** is attached at the second or inner side **54** to the first or outer side **46** of the second resistive element **13**. The third conductive portion **16** and the second resistive element **13** may be attached, connected, joined, bonded, or attached by various means such as welding, bonding, adhesives, ultrasonic bonding, soldering, brazing, or other methods that provide a low resistive connection, or any other acceptable connection means providing for electrical and/or thermal communication.

[0072] In the arrangement as shown in FIG. **1**, the second conductive portion **14** and the third conductive portion **16** are positioned on opposite sides of the second resistive element **13**.

[0073] Although the conductive portions **12**, **16** are illustrated as being a similar size and shape, the conductive portions **12**, **16** can be different in one aspect of the invention. For example, the size and shape of the conductive portions **12**, **16**, as well as the material forming the conductive portions **12**, **16** can vary. The conductive portions **12**, **16** are illustrated with four sides, but the number of sides for the conductive portions **12**, **16** can vary. These variations can be realized in any one or more of the other embodiments described herein.

[0074] Similarly, although the resistive elements **11**, **13** are illustrated as being a similar size and shape, the resistive elements **11**, **13** can be different in size, shape, or the materials comprising the resistive elements. For example, the size, dimension and/or shape of the resistive elements **11**, **13**, as well as the material forming the resistive elements **11**, **13** can vary. The resistive elements **11**, **13** are illustrated with four sides, but the number of sides for the resistive elements **11**, **13** can vary. These variations can be realized in any one or more of the other embodiments described herein.

[0075] In an aspect of the invention, the first conductive portion **12** may have a first opening **70** adjacent the first or outer side **18** of the first conductive portion **12**. The first opening **70** may comprise a hole completely through a portion of the first conductive portion **12** and may be generally circular in shape. The first opening **70** is positioned closer to the first side **18** of the first conductive portion **12**. The first opening **70** may function as a current connection hole or area. The current connection opening **70** is only one example of how a resistor may be attached to a current that is to be monitored and/or measured. Such openings may be omitted entirely.

[0076] A second opening **72** is formed adjacent the second or inner side **20** of the first conductive portion **12**. The second opening **72** may comprise a hole completely through a portion of the first conductive portion **12** and be generally oblong in shape, such as a curved oblong shape. The second opening **72** is positioned closer to the second side **20**. The second opening **72** functions as a TCR adjustment opening, which may also be referred to as a TCR compensation slot, as described in greater detail herein.

[0077] In an aspect of the invention, the third conductive portion **16** has a first opening **74** adjacent the first or outer side **56** of the third conductive portion **16**. The first opening **74** may comprise a hole completely through a portion of the third conductive portion **16** and may be generally circular in shape. The first opening **74** is positioned closer to the first side **56** of the third conductive portion **16**. The first opening **74** may function as a current connection hole or area. The current connection opening **74** is only one example of how a resistor may be attached to a current that is to be monitored and/or measured. Such openings may be omitted entirely.

[0078] The third conductive portion **16** includes a second opening **76** adjacent the second or inner side **54** of the third conductive portion **16**. The second opening **76** may comprise a hole completely through a portion of the third conductive portion **16** and be generally oblong in shape, such as a curved oblong shape. The second opening **76** is positioned closer to the second side **54**. The second opening **76** functions as a TCR adjustment opening, which may also be referred to as a TCR compensation slot, as described in greater detail herein.

[0079] In an aspect of the invention, the second conductive portion **14** includes a central opening **78** that may be generally centrally located through the second conductive portion **14**. The central opening **78** is generally oblong in shape, or may have a varied shape as further described herein. The central opening **78** functions as a TCR adjustment opening, which may also be referred to as a TCR compensation slot, as described in greater detail herein.

[0080] In an aspect of the invention, the first opening **70** of the first conductive portion **12** and the first opening **74** of the third conductive portion **16** have generally the same shape. In an aspect of the invention, the second opening **72** of the first conductive portion **12** and the second opening **76** of the third conductive portion **16** have the generally the same shape. In an aspect of the invention, the first conductive portion and the third conductive portions are mirrors of each other.

[0081] All of the openings (i.e. openings **70**, **72**, **74**, **76**, and **78**) of the resistor **10** may be aligned along a longitudinal axis **L1** of the resistor **10**. Alternatively, in different embodiments, certain

openings can be offset from each other.

[0082] As shown in FIG. 1, the openings **70, 72, 74, 76, and 78** are all arranged in, located within, or disposed in, an interior area of the resistor. As used herein, the term interior area with respect to the disclosed resistors means an area located away from the perimeter or outer edges of the resistor, or the perimeter outer edges of any components of the resistor, such as the resistive elements or conductive portions. The openings **70, 72, 74, 76, and 78** are therefore each preferably spaced away from outer edges, such as the lateral and longitudinal edges, or other portions of the perimeter or edges of the resistor **10**. Preferably, no portion of the openings **70, 72, 74, 76, and 78** intersect, contact or cross the outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **10**. In this way, the openings **70, 72, 74, 76, and 78** are surrounded on all sides by parts of the conductive portions, the resistive elements, or combinations of those.

[0083] Openings **70, 74** are illustrated as generally being circular in FIG. 1. However, a profile of the openings **70, 74** may vary in different aspects of the invention. The term profile is used herein to refer to the overall outline, size, shape, cross-section, orientation, or other physical characteristic. The openings **70, 74** may be identical or different from each other. In one aspect, at least one of the openings **70, 74** has an elongated, slotted, or non-circular profile. The orientation of the elongated or slotted profile of the openings **70, 74** may extend in the X-direction. In another aspect, the orientation of the elongated or slotted profile of the openings **70, 74** may extend in the Y-direction. The openings **70, 74** may not extend completely through the resistor in another aspect. In certain aspects, one or more of the openings **70, 74** may be omitted completely. These variations can be realized in any one or more of the other embodiments described herein.

[0084] All aspects of the openings **72, 76, 78** may also vary. For example, the openings **72, 76, 78** may each have a different profile, dimension, size, or shape. These may be adjusted or changed based on need or function. The openings **72, 76, 78** may include one or more through openings and one or more openings that do not extend completely through the conductive portions and/or the resistive elements. The openings **72, 76, 78** may be circular, elongated, slotted, non-circular, S-shaped, N-shaped, serpentine, or any other shape as may be selected. The orientation or direction in which any of the slotted or elongated openings **72, 76, 78** extend may vary, having differences in height, length, width, etc. In certain aspects, one or more of the openings **72, 76, 78** may be omitted. These variations can be realized in any one or more of the other embodiments described herein.

[0085] The openings in the conductive portions closest to the resistive elements (e.g., the TCR adjustment openings) create or define parts of the conductive portions that function as voltage sense terminal areas adjacent to the TCR adjustment openings. These areas may be connected to voltage sense pins, as further described herein. Such voltage sense terminal areas are smaller in size than the areas of the conductive portions that define the main current terminals.

[0086] At least one bracket or header assembly **80a, 80b** can be provided in one aspect, as shown in FIG. 1, which may be a header pin holding assembly as known in the art, and may be referred to as “brackets” or “header assemblies.” Brackets or header assemblies **80a, 80b, 180, 2180, 2280, 2380a, 2380b, 2480a, 2480b** are illustrated in phantom lines in some of the Figures for illustrative purposes only. Additional Figures, such as at least FIGS. **3, 4, 5, 6, 8, 9, 13A-13E, 17A-17M, 18, 19, 20** show aspects of brackets or header assemblies which are not specifically annotated or further described herein. One skilled in the art would understand that the brackets or header assemblies of any one of the Figures disclosed herein can be implemented and adapted to be used in any other one of the Figures.

[0087] The at least one bracket or header assembly **80a, 80b** comprises a brace **82** and pins **84**. In one aspect, the pins **84** are configured as mounting pins. As shown in FIG. 1, a first bracket or header assembly **80a** and a second bracket or header assembly **80b** can be provided. Two braces and four mounting pins **84** can be provided in one embodiment. In one aspect, one or more of the pins **84** can be sensing pins or ground pins. The bracket or header assemblies **80a, 80b** are

configured to attach and electrically connect the resistor **10** to another component.

[0088] In FIG. **1**, the openings **70** and **74** are current attachment points. Pins of the bracket or header assembly **80a** create a pair of voltage sense connections for first resistive element **11**, and the pins in the bracket or header assembly **80b** create a pair of voltage sense connections for the second resistive element **13**.

[0089] A resistor is therefore provided, in an aspect of the invention, with two distinct resistive elements which can match or not match in resistance value, and that in turn provide multiple independent voltage sense points for each of the elements in a manner represented but may also be configured in other formats for connection or arrangement or quantity of elements and/or pins. Aspects of the present invention also incorporate alternate methods of creating, and different arrangements of, slots, opening or holes in the conductive or terminal portions, to provide TCR compensation values and adjustments that expand or improve upon the methods shown and described in U.S. Pat. No. 8,878,643, the entire contents of which is incorporated by reference herein. The TCR compensation is achieved by directing the current around the sense points, causing a shadowing effect. Having the sense points in a low current density portion of the device allows for the control of the contribution that the conductive terminal material has on the TCR value of the resistor. The manner in which this feature is created allows for the positioning and attachment to, for example, mounting pins, in a way that allows for connection of the resistor to a surface created in a selected TCR opening or slot, allowing the connection to be placed in a more precise and controlled location. This isolation feature may or may not be incorporated depending on the specific requirements of the application, particularly for controlling, adjusting, or monitoring temperature coefficient of resistance.

[0090] The TCR improvement is achieved in the following manner. The TCR of copper is 3900 PPM/° C. while the resistive strip **11** may be 100 PPM/° C. The size, shape and distance of the TCR adjustment openings **72**, **76**, **78** with respect to the resistive elements **11**, **13** can be adjusted in length in the “Y” directions (e.g., transversely) and with respect to the “X” directions (e.g., longitudinally) by a distance of the pins **84** of header with respect to the resistive elements **11**, **13**. The adjustment of the “Y” dimension can be used to adjust the amount of isolation that the pins associated with the bracket or header assembly **80** experience which then changes the contribution to the TCR of the copper portions or strips and resistive elements. This adjustment to this contribution directly affects the TCR of the resistor seen (e.g., measured) in the voltage sensed at the pins of the bracket or header assembly **80**. This is true of any combination of voltage sense points, resistive strip and isolation holes on a resistor. The relationship of these features allows for the tuning of the TCR values with precision, and in an efficient manner.

[0091] FIG. **12** illustrates the current flux of a resistor **1210** including a first conductive portion **1212**, a second conductive portion **1214**, a third conductive portion **1216**, a first resistive element **1211**, and a second resistive element **1213**. The current flux is directed towards edges of the resistor **1210** and away from the sense points. The current flow in the X-axis is minimal between openings **1272**, **1276**, and **1278**.

[0092] As shown in FIG. **12**, the openings **1270**, **1272**, **1274**, **1276**, **1278** are all arranged in an interior area of the resistor **1210**. The openings **1270**, **1272**, **1274**, **1276**, **1278** are therefore each preferably spaced away from outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **1210**. No portion of the openings **1270**, **1272**, **1274**, **1276**, **1278** intersect, contact or cross the outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **1210**. In this way, the openings **1270**, **1272**, **1274**, **1276**, **1278** are surrounded on all sides by parts of the conductive portions, the resistive elements, or combinations of those.

[0093] In another aspect of the invention, shown in FIG. **2**, a resistor **110** is provided. The resistor **110** is generally formed of a resistive element **111** disposed between a first conductive portion **112** and a second conductive portion **114**.

[0094] As oriented in the depiction of FIG. **2**, the first conductive portion **112** has a first or outer

side **118** facing in a first direction **X1**, and a second or inner side **120** facing in a second direction **X2** which is a direction opposite the first direction **X1**, a third side **122** facing in a direction **Y1**, and a fourth side **124** facing in a direction **Y2** which is a direction opposite the direction **Y1**. The first conductive portion **112** may be formed generally as a plate or a strip or a bar. The first conductive portion **112** may be formed of a conductive metal. In an embodiment, the first conductive portion **112** is formed from copper (Cu) or a Cu alloy. Other conductive materials that may be used to form the first conductive portion **112** are, either alone or in combination, tin (Sn), aluminum (Al), silver (Ag), or gold (Au), with the alloy being plated or not plated with layers of nickel (Ni), Sn, or other similar materials.

[0095] The resistive element **111** has a first side **126** facing in the first direction **X1**, and a second side **128** facing in the second direction **X2**, a third side **130** facing in the direction **Y1**, and a fourth side **132** facing in the direction **Y2**. The resistive element **111** may be formed generally as a plate or a strip or a bar. The resistive element **111** may be formed of a resistive material, or combinations of materials including resistive materials. In an embodiment, the resistive element **111** is formed from or comprises alloys of copper-nickel-manganese (CuNiMn), copper manganese tin (CuMnSn), copper nickel (CuNi), nickel-chromium-aluminum (NiCrAl), or nickel-chromium (NiCr), or other alloys known to those of skill in the art acceptable for use as a resistive element as part of a shunt resistor. Other resistive materials that may be used to form the resistive element **111** are, either alone or in combination, Cu, Ni, Mn, Cr, Al, and Fe.

[0096] The first conductive portion **112** is attached at the second or inner side **120** to the first or outer side **126** of the resistive element **111**. The first conductive portion **112** and the resistive element **111** may be attached, connected, joined, bonded, or attached by various means such as welding, bonding, adhesives, ultrasonic bonding, soldering, brazing, or other methods that provide a low resistive connection, or any other acceptable connection means providing for electrical and/or thermal communication. The first conductive portion **112** is generally located at a first side **134** or first portion of the resistor **110**.

[0097] As oriented in the depiction of FIG. 2, the second conductive portion **114** has a first side **136** facing in the second direction **X2**, and a second side **138** facing in the first direction **X1**, a third side **140** facing in the direction **Y1**, and a fourth side **142** facing in the direction **Y2**. The second conductive portion **114** may be formed generally as a plate or a strip or a bar. The second conductive portion **114** may be formed of a conductive metal. In an embodiment, the second conductive portion **114** is formed from copper (Cu) or a Cu alloy. Other conductive materials that may be used to form the second conductive portion **114** are, either alone or in combination, tin (Sn), aluminum (Al), silver (Ag), or gold (Au), with the alloy being plated or not plated with layers of nickel (Ni), Sn, or other similar materials. The second conductive portion **114** is generally located or positioned adjacent a second side **144** or second portion of the resistor **110**.

[0098] The second conductive portion **114** is attached at the second side **138** to the second side **128** of the resistive element **111**. Accordingly, the first conductive portion **112** and the second conductive portion **114** are oriented on opposite sides of the resistive element **111**. The second conductive portion **114** and the resistive element **111** may be connected, joined, bonded, or attached by various means such as welding, bonding, adhesives, ultrasonic bonding, soldering, brazing, or other methods that provide a low resistive connection, or any other acceptable connection means providing for electrical and/or thermal communication.

[0099] In an aspect of the invention, the first conductive portion **112** has a first opening **170** adjacent the first or outer side **118** of the first conductive portion **112**, and a second opening **172** adjacent the second or inner side **120** of the first conductive portion **112**. The first opening **170** may comprise a hole completely through a portion of the first conductive portion **112** and may be generally circular in shape. The first opening **170** is positioned closer to the first side **118** of the first conductive portion **112**. The first opening **170** may function as a current connection hole or area. The current connection opening **170** is only one example of how a resistor may be attached to

a current that is to be monitored and/or measured. Such openings may be omitted entirely.

[0100] The second opening **172** may comprise a hole completely through a portion of the first conductive portion **112** and be generally oblong in shape, such as a curved oblong shape. The second opening **172** is positioned closer to the second side **120**. The second opening **172** functions as a TCR adjustment opening, which may also be referred to as a TCR compensation slot, as described in greater detail herein.

[0101] In an aspect of the invention, the second conductive portion **114** has a first opening **174** adjacent the first or outer side **136** of the second conductive portion **114**, and a second opening **176** adjacent the second or inner side **138** of the second conductive portion **114**. The first opening **174** may comprise a hole completely through a portion of the second conductive portion **114** and may be generally circular in shape. The second opening **176** may comprise a hole completely through a portion of the second conductive portion **114** and be generally oblong in shape, such as a curved oblong shape. The first opening **174** is positioned closer to the first side **136** of the second conductive portion **114** and the second opening **176** is positioned closer to the second side **138**.

[0102] In an aspect of the invention, the first opening **170** of the first conductive portion **112** and the first opening **174** of the second conductive portion **114** have generally the same shape. In an aspect of the invention, the second opening **172** of the first conductive portion **112** and the second opening **176** of the second conductive portion **114** have the generally the same shape.

[0103] All of the openings (i.e. openings **170**, **172**, **174**, and **176**) of the resistor **110** may be aligned along a longitudinal axis L2 of the resistor **110**. Alternatively, in different embodiments, certain openings can be offset from each other.

[0104] As shown in FIG. 2, the openings **170**, **172**, **174**, and **176** are all arranged in an interior area of the resistor **110**. The openings **170**, **172**, **174**, and **176** are each spaced away from outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **110**. No portion of the openings **170**, **172**, **174**, and **176** intersect, contact or cross the outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **110**. In this way, the openings **170**, **172**, **174**, and **176** are surrounded on all sides by parts of the conductive portions, the resistive elements, or combinations of those.

[0105] As shown in FIG. 2, a bracket or header assembly **180** can be provided for attaching the resistor **110** to another component, such as a header pin. The bracket or header assembly **180** is similar to the bracket or header assembly **80** of FIG. 1, and also includes braces and mounting pins.

[0106] As shown in FIG. 2, the openings **170** and **174** are the current connections, the upper pins of the bracket or header assembly **180** are one pair of voltage senses and the lower pair of pins of the bracket or header assembly **180** are the other voltage sense pair.

[0107] Openings **170**, **174** are illustrated as generally being circular in FIG. 2. However, a profile of the openings **170**, **174** may vary in different aspects of the invention. The openings **170**, **174** may be identical or different from each other. In one aspect, at least one of the openings **170**, **174** has an elongated, slotted, or non-circular profile. The orientation of the elongated or slotted profile of the openings **170**, **174** may extend in the X-direction. In another aspect, the orientation of the elongated or slotted profile of the openings **170**, **174** may extend in the Y-direction. The openings **170**, **174** may not extend completely through the resistor in another aspect. In certain aspects, one or more of the openings **170**, **174** may be omitted completely. These variations can be realized in any one or more of the other embodiments described herein.

[0108] All aspects of the openings **172**, **176** may also vary. For example, the openings **172**, **176** may each have a different profile, dimension, size, or shape. These may be adjusted or changed based on need or function. The openings **172**, **176** may include one or more through openings and one or more openings that do not extend completely through the conductive portions and/or the resistive elements. The openings **172**, **176** may be circular, elongated, slotted, non-circular, S-shaped, N-shaped, serpentine, or any other shape as may be selected. The orientation or direction in which any of the slotted or elongated openings **172**, **176** extend may vary, having differences in

height, length, width, etc. In certain aspects, one or more of the openings **172**, **176** may be omitted. These variations can be realized in any one or more of the other embodiments described herein.

[0109] In another aspect of the invention shown in FIG. **3**, a resistor **210** is disclosed that is generally formed of a first resistive element **211** disposed between a first conductive portion **212** and a second conductive portion **214**, and a second resistive element **213** disposed between the second conductive portion **214** and a third conductive portion **216**. The resistor **210** is similar to the resistor **10** in that both resistors include two resistive elements **11**, **13**, **211**, and **213**.

[0110] The resistor **210** generally has a curved profile as opposed to the linear profile of resistors **10** and **110**. In one aspect, the resistor **210** has a U-shaped profile or a “horseshoe” profile. The first conductive portion **212** and the third conductive portion **216** have a linear or straight profile and the second conductive portion **214** has a curved or U-shaped profile. The second conductive portion **214** may have a square profile in one aspect. One of ordinary skill in the art would understand that the shape of the conductive portions **212**, **214**, **216** can vary.

[0111] The first resistive element **211** has a first or outer side **226** facing in the second direction **X2**, and a second or inner side **228** facing in the first direction **X1**, a third side **230** facing in the direction **Y1**, and a fourth side **232** facing in the direction **Y2**. The first resistive element **211** may be formed generally as a plate or a strip or a bar. The first resistive element **211** may be formed of a resistive material, or combinations of materials including resistive materials. In an embodiment, the first resistive element **211** is formed from or comprises alloys of copper-nickel-manganese (CuNiMn), copper manganese tin (CuMnSn), copper nickel (CuNi), nickel-chromium-aluminum (NiCrAl), or nickel-chromium (NiCr), or other alloys known to those of skill in the art acceptable for use as a resistive element as part of a shunt resistor. Other resistive materials that may be used to form the first resistive element **211** are, either alone or in combination, Cu, Ni, Mn, Cr, Al, and Fe.

[0112] The first conductive portion **212** has a first or outer side **218** facing in the second direction **X2**, and a second or inner side **220** facing in the first direction **X1**, a third side **222** facing in the direction **Y1**, and a fourth side **224** facing in the direction **Y2**. The first conductive portion **212** may be formed generally as a plate or a strip or a bar. The first conductive portion **212** may be formed of a conductive metal. In an embodiment, the first conductive portion **212** is formed from copper (Cu) or a Cu alloy. Other conductive materials that may be used to form the first conductive portion **12** are, either alone or in combination, Sn, Al, Ag, or Au, with the alloy being plated or not plated with layers of Ni, Sn, or other similar materials.

[0113] The first conductive portion **212** is attached at the second or inner side **220** to the first or outer side **226** of the first resistive element **211**. The first conductive portion **212** and the first resistive element **211** may be connected, joined, bonded, or attached by various means such as welding, bonding, adhesives, ultrasonic bonding, soldering, brazing, or other methods that provide a low resistive connection, or any other acceptable connection means providing for electrical and/or thermal communication. The first conductive portion **212** is generally located at a first side **234** or first portion of the resistor **210**.

[0114] As oriented in the depiction of FIG. **3**, the second conductive portion **214** has a first side **236** facing in the second direction **X2**, and a second side **238** also facing in the second direction **X2**, an inner curved side **240** and an opposite, outer curved side **242**. The second conductive portion **214** may be formed generally as a plate or a strip or a bar. In one aspect, as shown in FIG. **3**, the second conductive portion **214** has a curved profile. In one aspect, the second conductive portion **214** has a U-shaped profile. The second conductive portion **214** may be formed of a conductive metal. In an embodiment, the second conductive portion **214** is formed from copper (Cu) or a Cu alloy. Other conductive materials that may be used to form the second conductive portion **214** are, either alone or in combination, Sn, Al, Ag, or Au, with the alloy being plated or not plated with layers of Ni, Sn, or other similar materials. The second conductive portion **214** is generally located or positioned adjacent at a central location or midpoint of the resistor **210**.

[0115] The second conductive portion **214** is attached at the first side **236** to the second or inner side **228** of the first resistive element **211**. Accordingly, the first conductive portion **212** and the second conductive portion **214** are oriented on opposite sides of the first resistive element **211**. The second conductive portion **214** and the first resistive element **211** may be connected, joined, bonded, or attached by various means such as welding, bonding, adhesives, ultrasonic bonding, soldering, brazing, or other methods that provide a low resistive connection, or any other acceptable connection means providing for electrical and/or thermal communication.

[0116] The second resistive element **213** has a first or outer side **246** facing in the second direction **X2**, and a second or inner side **248** facing in the first direction **X1**, a third side **250** facing in the direction **Y1**, and a fourth side **252** facing in the direction **Y2**. The second resistive element **213** may be formed generally as a plate or a strip or a bar. The second resistive element **213** may be formed of a resistive material, or combinations of materials including resistive materials. In an embodiment, the second resistive element **213** is formed from or comprises alloys of copper-nickel-manganese (CuNiMn), copper manganese tin (CuMnSn), copper nickel (CuNi), nickel-chromium-aluminum (NiCrAl), or nickel-chromium (NiCr), or other alloys known to those of skill in the art acceptable for use as a resistive element as part of a shunt resistor. Other resistive materials that may be used to form the second resistive element **213** are, either alone or in combination, Cu, Ni, Mn, Cr, Al, and Fe.

[0117] The second conductive portion **214** is attached at the second or inner side **238** to the second or inner side **248** of the second resistive element **213**. The second conductive portion **214** and the second resistive element **213** may be connected, joined, bonded, or attached by various means such as welding, bonding, adhesives, ultrasonic bonding, soldering, brazing, or other methods that provide a low resistive connection, or any other acceptable connection means providing for electrical and/or thermal communication.

[0118] In the arrangement as shown in FIG. 3, the first resistive element **211** and second resistive element **213** are positioned on opposite sides of the second conductive portion **214**.

[0119] As oriented in the depiction of FIG. 3, the third conductive portion **216** has an inner or second side **254** facing in the first direction **X1**, and a first or outer side **256** facing in the second direction **X2**, a third side **258** facing in the direction **Y1**, and a fourth side **260** facing in the direction **Y2**. The third conductive portion **216** may be formed generally as a plate or a strip or a bar. The third conductive portion **216** may be formed of a conductive metal. In an embodiment, the third conductive portion **216** is formed from copper (Cu) or a Cu alloy. Other conductive materials that may be used to form the third conductive portion **216** are, either alone or in combination, Sn, Al, Ag, or Au, with the alloy being plated or not plated with layers of Ni, Sn, or other similar materials. The second conductive portion **214** is generally located or positioned adjacent at a second side **262** or second portion of the resistor **210**.

[0120] The third conductive portion **216** is attached at the second or inner side **254** to the first or outer side **246** of the second resistive element **213**. The third conductive portion **216** and the second resistive element **213** may be connected, joined, bonded, or attached by various means such as welding, bonding, adhesives, ultrasonic bonding, soldering, brazing, or other methods that provide a low resistive connection, or any other acceptable connection means providing for electrical and/or thermal communication.

[0121] In the arrangement as shown in FIG. 3, the second conductive portion **214** and the third conductive portion **216** are positioned on opposite sides of the second resistive element **213**. The second conductive portion **214** may also be attached to a heat sink to dissipate heat and improve the long-term stability of the part and reduce the resistive element temperature in operation.

[0122] In an aspect of the invention, the first conductive portion **212** has a first opening **270** adjacent the first or outer side **218** of the first conductive portion **212**, and a second opening **272** adjacent the second or inner side **220** of the first conductive portion **212**. The first opening **270** may comprise a hole completely through a portion of the first conductive portion **212** and may be

generally circular in shape. The second opening **272** may comprise a hole completely through a portion of the first conductive portion **212** and be generally oblong in shape. Preferably, the second opening **272** is a generally C-shaped opening functioning as a TCR adjustment slot. As used herein, the term “C-shaped” may designate, define, refer to, or capture, multiple shapes, which may be curved shapes, such as, but not limited to, a curved opening, or an opening tracing or in the shape of an arc, or being arc-shaped, or arch-shaped, or a parabolic shape. The C-shaped opening has a curved portion, such as an outwardly curved or outwardly curving portion, curving toward the first side **218** of the first conductive portion **212**, with a concave side facing the first resistive element **211**. A first dimple, indentation or cut-out **272a** may be formed as a semi-circular shape at a first side of the second opening **272**. A second dimple, indentation or cut-out **272b** may be formed as a semi-circular shape at a second side of the second opening **272**. These cut-outs provide areas for pin connections and placement. The first opening **270** is positioned closer to the first side **218** of the first conductive portion **212** and the second opening **272** is positioned closer to the second side **220**.

[0123] The first opening **270** and the second opening **274** comprise or provide for current connections and the pins closest to the first resistive element **211** of cut-out **272b** and cut-out **278a** comprise or provide for voltage connections for the resistive element **211**. Thus, first opening **270** and second opening **274** are positioned adjacent what can be referred to as main current terminals of the conductive portions.

[0124] It is appreciated that the TCR adjustment openings in any disclosed embodiment form, define or comprise voltage sense terminals in areas of the conductive portions adjacent to the TCR adjustment openings. Accordingly, the designs shown and described herein provide for a wide range of positioning choices for the pins along or within the TCR adjustment openings to allow for voltage sensing at selected positions.

[0125] The pins contacting or otherwise connected to first cut-out **279a** and second cut-out **276b** closest to side **252** comprise or provide for the voltage sense pins for the resistive element **213**. Thus, cut-out **279a** and cut-out **276b** are positioned adjacent what can be referred to as voltage sense terminals of the conductive portions.

[0126] In an aspect of the invention, the third conductive portion **216** has a first opening **274** adjacent the first or outer side **256** of the third conductive portion **216**, and a second opening **276** adjacent the second or inner side **254** of the third conductive portion **216**. The first opening **274** may comprise a hole completely through a portion of the third conductive portion **216** and may be generally circular in shape. The second opening **276** may comprise a hole completely through a portion of the third conductive portion **216** and be generally oblong in shape. The first opening **274** is positioned closer to the first side **256** of the third conductive portion **216** and the second opening **276** is positioned closer to the second side **254**. The second opening **276** is preferably a generally C-shaped opening functioning as a TCR adjustment slot. The C-shaped opening has a curved portion, such as an outwardly curved or outwardly curving portion, curving toward the first side **256** of the third conductive portion **216** with a concave side facing the second resistive element **213**. That is, the C-shaped opening curves outwardly away from a central portion of the resistor. A first dimple, indentation or cut-out **276a** may be formed as a semi-circular shape at a first side of the second opening **276**. A second dimple, indentation or cut-out **276b** may be formed as a semi-circular shape at a second side of the second opening **276**. These cut-outs provide areas for pin connections and placement.

[0127] In an aspect of the invention, the second conductive portion **214** includes a first opening **278** that is generally located adjacent to the first side **236** and a second opening **279** generally located adjacent to the second side **238**. The openings **278**, **279** may be generally oblong in shape. The openings **278**, **279**, as shown in FIG. 3, are generally C-shaped openings functioning as TCR adjustment openings or slots. The C-shaped opening has a curved portion, such as an outwardly curved or outwardly curving portion, curving toward the outer curved side **242** of the second

conductive portion **214** with a concave side facing the first and second resistive elements **211**, **213**, respectively. A first dimple, indentation or cut-out **278a**, **279a** may be formed as a semi-circular shape at a first side of the openings **278**, **279**. A second dimple, indentation or cut-out **278b**, **279b** may be formed as a semi-circular shape at a second side of the openings **278**, **279**. These cut-outs provide areas for pin connections and placement. Soldering pins to a surface between the openings and the resistive elements can also provide a voltage sense point. It is appreciated that various external connections or sense leads may be connected to the voltage sense areas of the resistors disclosed herein.

[0128] In an aspect of the invention, the first opening **270** of the first conductive portion **212** and the first opening **274** of the third conductive portion **216** have generally the same shape. However, those openings can be selectively adjusted to have different shapes and to be of different sizes.

[0129] In an aspect of the invention, the second opening **272** of the first conductive portion **212**, the second opening **276** of the third conductive portion **216**, and the opening **278**, **279** of the second conductive portion **214** have the generally the same shape. However, those openings can be selectively adjusted to have different shapes and to be of different sizes.

[0130] Openings **270**, **274** are illustrated as generally being circular in FIG. **3**. However, a profile of the openings **270**, **274** may vary in different aspects of the invention. The openings **270**, **274** may be identical or different from each other. In one aspect, at least one of the openings **270**, **274** has an elongated, slotted, or non-circular profile. The orientation of the elongated or slotted profile of the openings **270**, **274** may extend in the X-direction. In another aspect, the orientation of the elongated or slotted profile of the openings **270**, **274** may extend in the Y-direction. The openings **270**, **274** may not extend completely through the resistor in another aspect. In certain aspects, one or more of the openings **270**, **274** may be omitted completely. These variations can be realized in any one or more of the other embodiments described herein.

[0131] All aspects of the openings **272**, **276**, **278**, **279** may also vary. For example, the openings **272**, **276**, **278**, **279** may each have a different profile, dimension, size, or shape. These may be adjusted or changed based on need or function. The openings **272**, **276**, **278**, **279** may include one or more through openings and one or more openings that do not extend completely through the conductive portions and/or the resistive elements. The openings **272**, **276**, **278**, **279** may be circular, elongated, slotted, non-circular, S-shaped, N-shaped, serpentine, or any other shape as may be selected. The orientation or direction in which any of the slotted or elongated openings **272**, **276**, **278**, **279** extend may vary, having differences in height, length, width, etc. In certain aspects, one or more of the openings **272**, **276**, **278**, **279** may be omitted. These variations can be realized in any one or more of the other embodiments described herein.

[0132] As shown in FIG. **3**, the openings **272**, **276**, **278**, and **279** are all arranged in an interior area of the resistor **210**. The openings **272**, **276**, **278**, and **279** are each spaced away from outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **210**. No portion of the openings **272**, **276**, **278**, and **279** intersect, contact or cross the outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **210**. In this way, the openings **272**, **276**, **278**, and **279** are surrounded on all sides by parts of the conductive portions, the resistive elements, or combinations of those.

[0133] As shown in FIG. **3**, a bracket or header assembly **280** can be provided for attaching the resistor **210** to another component. The bracket or header assembly **280** is similar to the bracket or header assembly **80** of FIG. **1**, and also includes braces and mounting pins.

[0134] FIG. **4** shows a resistor **310** generally formed of a first resistive element **311** disposed between a first conductive portion **312** and a second conductive portion **314**, and a second resistive element **313** disposed between the second conductive portion **314** and a third conductive portion **316**.

[0135] As oriented in the depiction of FIG. **4**, the first conductive portion **312** has a first or outer side **318** facing in a first direction **X1**, and a second or inner side **320** facing in a second direction

X2 which is a direction opposite the first direction X1, a third side 322 facing in a direction Y1, and a fourth side 324 facing in a direction Y2 which is a direction opposite the direction Y1.

[0136] The first conductive portion 312 may be formed generally as a plate or a strip or a bar. The first conductive portion 312 may be formed of a conductive metal. In an embodiment, the first conductive portion 312 is formed from copper (Cu) or a Cu alloy. Other conductive materials that may be used to form the first conductive portion 12 are, either alone or in combination, Sn, Al, Ag, or Au, with the alloy being plated or not plated with layers of Ni, Sn, or other similar materials.

[0137] The first resistive element 311 has a first or outer side 326 facing in the first direction X1, and a second or inner side 328 facing in the second direction X2, a third side 330 facing in the direction Y1, and a fourth side 332 facing in the direction Y2. The first resistive element 311 may be formed generally as a plate or a strip or a bar. The first resistive element 311 may be formed of a resistive material, or combinations of materials including resistive materials. In an embodiment, the first resistive element 311 is formed from or comprises alloys of copper-nickel-manganese (CuNiMn), copper manganese tin (CuMnSn), copper nickel (CuNi), nickel-chromium-aluminum (NiCrAl), or nickel-chromium (NiCr), or other alloys known to those of skill in the art acceptable for use as a resistive element as part of a shunt resistor. Other resistive materials that may be used to form the first resistive element 311 are, either alone or in combination, Cu, Ni, Mn, Cr, Al, and Fe.

[0138] The first conductive portion 312 is attached at the second or inner side 320 to the first or outer side 326 of the first resistive element 311. The first conductive portion 312 and the first resistive element 311 may be connected, joined, bonded, or attached by various means such as welding, bonding, adhesives, ultrasonic bonding, soldering, brazing, or other methods that provide a low resistive connection, or any other acceptable connection means providing for electrical and/or thermal communication. The first conductive portion 312 is generally located at a first side 334 or first portion of the resistor 310.

[0139] As oriented in the depiction of FIG. 4, the second conductive portion 314 has a first side 336 facing in the first direction X1, and a second side 338 facing in the second direction X2, a third side 340 facing in the direction Y1, and a fourth side 342 facing in the direction Y2. The second conductive portion 314 may be formed generally as a plate or a strip or a bar. The second conductive portion 314 may be formed of a conductive metal. In an embodiment, the second conductive portion 314 is formed from copper (Cu) or a Cu alloy. Other conductive materials that may be used to form the second conductive portion 314 are, either alone or in combination, Sn, Al, Ag, or Au, with the alloy being plated or not plated with layers of Ni, Sn, or other similar materials. The second conductive portion 314 is generally located or positioned adjacent at a central location or midpoint of the resistor 310.

[0140] The second conductive portion 314 is attached at the first side 336 to the second or inner side 328 of the first resistive element 311. Accordingly, the first conductive portion 312 and the second conductive portion 314 are oriented on opposite sides of the first resistive element 311. The second conductive portion 314 and the first resistive element 311 may be connected, joined, bonded, or attached by various means such as welding, bonding, adhesives, ultrasonic bonding, soldering, brazing, or other methods that provide a low resistive connection, or any other acceptable connection means providing for electrical and/or thermal communication.

[0141] The second resistive element 313 has a first or outer side 346 facing in the second direction X2, and a second or inner side 348 facing in the first direction X1, a third side 350 facing in the direction Y1, and a fourth side 352 facing in the direction Y2. The second resistive element 313 may be formed generally as a plate or a strip or a bar. The second resistive element 313 may be formed of a resistive material, or combinations of materials including resistive materials. In an embodiment, the second resistive element 313 is formed from or comprises alloys of copper-nickel-manganese (CuNiMn), copper manganese tin (CuMnSn), copper nickel (CuNi), nickel-chromium-aluminum (NiCrAl), or nickel-chromium (NiCr), or other alloys known to those of skill in the art

acceptable for use as a resistive element as part of a shunt resistor. Other resistive materials that may be used to form the second resistive element **313** are, either alone or in combination, Cu, Ni, Mn, Cr, Al, and Fe.

[0142] The second conductive portion **314** is attached at the second or inner side **338** to the second or inner side **348** of the second resistive element **313**. The second conductive portion **314** and the second resistive element **313** may be connected, joined, bonded, or attached by various means such as welding, bonding, adhesives, ultrasonic bonding, soldering, brazing, or other methods that provide a low resistive connection, or any other acceptable connection means providing for electrical and/or thermal communication.

[0143] In the arrangement as shown in FIG. 4, the first resistive element **311** and second resistive element **313** are positioned on opposite sides of the second conductive portion **314**.

[0144] As oriented in the depiction of FIG. 4, the third conductive portion **316** has an inner or second side **354** facing in the first direction X1, and a first or outer side **356** facing in the second direction X2, a third side **358** facing in the direction Y1, and a fourth side **360** facing in the direction Y2. The third conductive portion **316** may be formed generally as a plate or a strip or a bar. The third conductive portion **316** may be formed of a conductive metal. In an embodiment, the third conductive portion **316** is formed from copper (Cu) or a Cu alloy. Other conductive materials that may be used to form the third conductive portion **16** are, either alone or in combination, Sn, Al, Ag, or Au, with the alloy being plated or not plated with layers of Ni, Sn, or other similar materials. The second conductive portion **314** is generally located or positioned adjacent at a second side **362** or second portion of the resistor **310**.

[0145] The third conductive portion **316** is attached at the second or inner side **354** to the first or outer side **346** of the second resistive element **313**. The third conductive portion **316** and the second resistive element **313** may be connected, joined, bonded, or attached by various means such as welding, bonding, adhesives, ultrasonic bonding, soldering, brazing, or other methods that provide a low resistive connection, or any other acceptable connection means providing for electrical and/or thermal communication.

[0146] In the arrangement as shown in FIG. 4, the second conductive portion **314** and the third conductive portion **316** are positioned on opposite sides of the second resistive element **313**.

[0147] In an aspect of the invention, the first conductive portion **312** has a first opening **370** adjacent the first or outer side **318** of the first conductive portion **312**, and a second opening **372** adjacent the second or inner side **320** of the first conductive portion **312**. The first opening **370** may comprise a hole completely through a portion of the first conductive portion **312** and may be generally circular in shape. The second opening **372** may comprise a hole completely through a portion of the first conductive portion **312** and be generally curved in shape. The first opening **370** is positioned closer to the first side **318** of the first conductive portion **312** and the second opening **372** is positioned closer to the second side **320**. The second opening **372** is preferably a generally C-shaped opening functioning as a TCR adjustment slot. The C-shaped opening has a curved portion, such as an outwardly curved or outwardly curving portion, curving toward the first side **318** of the first conductive portion **312** with a concave side facing the first resistive element **311**. Additional features of the opening **372**, including dimple, indentations or cut-outs are described below.

[0148] In an aspect of the invention, the third conductive portion **316** has a first opening **374** adjacent the first or outer side **356** of the third conductive portion **316**, and a second opening **376** adjacent the second or inner side **354** of the third conductive portion **316**. The first opening **374** may comprise a hole completely through a portion of the third conductive portion **316** and may be generally circular in shape. The second opening **376** may comprise a hole completely through a portion of the third conductive portion **316** and be generally curved in shape. The first opening **374** is positioned closer to the first side **356** of the third conductive portion **316** and the second opening **376** is positioned closer to the second side **354**. The second opening **376** is preferably a generally

C-shaped opening functioning as a TCR adjustment slot. The C-shaped opening has a curved portion, such as an outwardly curved or outwardly curving portion, curving toward the first side **356** of the third conductive portion **316** with a concave side facing the second resistive element **313**. That is, the C-shaped opening curves outwardly away from a central portion of the resistor. Additional features of the opening **376**, including dimple, indentations or cut-outs are described below.

[0149] The openings **372**, **376** may arc or curve along the Y1-Y2 direction (transversely) or may arc or curve along the X1-X2 direction (longitudinally). As shown in FIG. 4, the openings **372**, **376** may include a first generally centrally located dimple, indentation or cut-out **372a**, **376a** providing for a generally semicircular opening extending in the X1 direction. This dimple, indentation or cut-out provides a first pin connection area. The openings **372**, **376** may include a second generally centrally located dimple, indentation or cut-out **372b**, **376b** providing for a generally semicircular opening extending in the X2 direction, which provides for a second pin connection area. The cut-outs **372a**, **372b**, **376a**, **376b** may lie generally along or be generally aligned along a longitudinal axis of the conductive portions.

[0150] The openings **370** and **374** are the current connections. The pins at cut-out **372b** and the pin at cut-out **378b** closest to **372b** are the voltage sense points for resistive element **311**, and the pins at cut-out **376a** and cut-out **378b** closest to the pin at cut-out **376a** are the voltage sense points for resistive element **313**.

[0151] In an aspect of the invention, the second conductive portion **314** includes a central opening **378** that is generally centrally located through the second conductive portion **314**. The central opening **378** may have a “dog bone” shape or barbell shape along its transverse length (in the Y1 to Y2 direction), having a larger portion **378a** at its upper area, and smaller portion **378b** at its central or middle area, and a larger portion **378c** at its lower area. The central opening **378** could also be rotated 90 degrees from the position shown in FIG. 4 in another variation. In one aspect, opening **372** is a first TCR adjustment opening, opening **376** is a second TCR adjustment opening, and opening **378** is a third TCR adjustment opening. The first and second TCR adjustment openings **372**, **376** each generally have a C-shaped profile and are mirror images and are symmetrical with each other, in one aspect. The third TCR adjustment opening **378** has a different profile than the first and second TCR adjustment openings **372**, **376**, in one aspect. In one aspect, the first and second TCR adjustment openings **372**, **376** each include a base region having an oblong or pill shaped profile and two curved arms extending on opposite sides of the oblong or pill shaped base region. In one aspect, the third TCR adjustment opening **378** includes three oblong or pill shaped regions connected to each other. In one aspect, each of the TCR adjustment openings **372**, **376**, **378** are symmetrical about the longitudinal axis L3. Each of the TCR adjustment openings **372**, **376**, **378** may be intersected by the longitudinal axis L3 at a respective midpoint.

[0152] In an aspect of the invention, the first opening **370** of the first conductive portion **312** and the first opening **374** of the third conductive portion **316** have generally the same shape. In an aspect of the invention, the second opening **372** of the first conductive portion **312**, the second opening **376** of the third conductive portion **316** have the generally the same shape.

[0153] All of the openings (i.e. openings **370**, **372**, **374**, **376**, and **378**) of the resistor **10** may be aligned along a longitudinal axis L3 of the resistor **310**. Alternatively, in different embodiments, certain openings can be offset from each other.

[0154] Although a longitudinal axis L1, L2, L3 is only explicitly illustrated in FIGS. 1, 2, and 4, respectively, one skilled in the art would understand that a longitudinal axis is defined in the longitudinal direction across each resistor illustrated in the Figures.

[0155] Openings **370**, **374** are illustrated as generally being circular in FIG. 4. However, a profile of the openings **370**, **374** may vary in different aspects of the invention. The term profile is used herein to refer to the overall outline, size, shape, cross-section, orientation, or other physical characteristic. The openings **370**, **374** may be identical or different from each other. In one aspect,

at least one of the openings **370**, **374** has an elongated, slotted, or non-circular profile. The orientation of the elongated or slotted profile of the openings **370**, **374** may extend in the X-direction. In another aspect, the orientation of the elongated or slotted profile of the openings **370**, **374** may extend in the Y-direction. The openings **370**, **374** may not extend completely through the resistor in another aspect. In certain aspects, one or more of the openings **370**, **374** may be omitted completely. These variations can be realized in any one or more of the other embodiments described herein.

[0156] All aspects of the openings **372**, **376**, **378** may also vary. For example, the openings **372**, **376**, **378** may each have a different profile, dimension, size, or shape. These may be adjusted or changed based on need or function. The openings **372**, **376**, **378** may include one or more through openings and one or more openings that do not extend completely through the conductive portions and/or the resistive elements. The openings **372**, **376**, **378** may be circular, elongated, slotted, non-circular, S-shaped, N-shaped, serpentine, or any other shape as may be selected. The orientation or direction in which any of the slotted or elongated openings **372**, **376**, **378** extend may vary, having differences in height, length, width, etc. In certain aspects, one or more of the openings **372**, **376**, **378** may be omitted. These variations can be realized in any one or more of the other embodiments described herein.

[0157] As shown in FIG. 4, the openings **370**, **372**, **374**, **376**, and **378** are all arranged in an interior area of the resistor **310**. The openings **370**, **372**, **374**, **376**, and **378** are each spaced away from outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **310**. No portion of the openings **370**, **372**, **374**, **376**, and **378** intersect, contact or cross the outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **310**. In this way, the openings **370**, **372**, **374**, **376**, and **378** are surrounded on all sides by parts of the conductive portions, the resistive elements, or combinations of those.

[0158] At least one bracket or header assembly **380** can be provided in one aspect, as shown in FIG. 4. The at least one bracket or header assembly **380** comprises a brace and mounting pins.

[0159] FIG. 18 illustrates a resistor **1810** that is similar to the resistor **310** of FIG. 4. Resistor **1810** is generally formed of a first resistive element **1811** disposed between a first conductive portion **1812** and a second conductive portion **1814**, and a second resistive element **1813** disposed between the second conductive portion **1814** and a third conductive portion **1816**. The resistor **1810** includes first, second, and third TCR adjustment slots **1872**, **1876**, **1878**, as well as a first and second outer openings **1870**, **1874**. The configuration, structure, function, and other characteristics of all elements of resistor **1810** in FIG. 18 are similar to the corresponding elements of resistor **310** in FIG. 4.

[0160] As shown in FIG. 18, the openings **1870**, **1872**, **1874**, **1876**, and **1878** are all arranged in an interior area of the resistor **1810**. The openings **1870**, **1872**, **1874**, **1876**, and **1878** are each spaced away from outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **1810**. No portion of the openings **1870**, **1872**, **1874**, **1876**, and **1878** intersect, contact or cross the outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **1810**. In this way, the openings **1870**, **1872**, **1874**, **1876**, and **1878** are surrounded on all sides by parts of the conductive portions, the resistive elements, or combinations of those.

[0161] FIG. 5 illustrates another embodiment of a resistor **410**. The resistor **410** is generally formed of a resistive element **411** disposed between a first conductive portion **412** and a second conductive portion **414**. The resistor **410** is similar to the resistor **110** of FIG. 2 and has the same function and structure as resistor **110** unless otherwise specified. As shown in FIG. 5, C-shaped TCR adjustment slots **472**, **476** and a horizontally or longitudinally positioned “dog bone” shaped or barbell shaped slot **478** that extends through the first conductive portion **412**, the second conductive portion **414**, and the resistive element **411**. In one aspect, indentations are defined on edges of the resistive element **411**. These indentations can be provided at varying depths, as well as other profiles, in order to provide adjustment of resistance values for the resistor. Any of the resistors disclosed

herein may include indentations or other modifications to the profiles of the resistive elements in order to adjust the resistance values of the resistors. In one aspect, the indentations adjust each section of the resistive element relative to a desired value. The outer circular openings in FIG. 5 are similar to the openings 70, 74 of FIG. 1. The upper pin of end 478a and the upper pin of end 478b are the voltage sense connections of 411. The lower pins of end 478a and the lower pin of end 478b are the voltage sense connections for element 411. Slot 478 acts as a further TCR adjustment opening or slot.

[0162] The C-shaped TCR adjustment slots 472, 476 are configured to curve about the larger end portions of the “dog bone” shaped or barbell shaped slot 478. The slots 472, 476 have a generally C-shaped or curved or arced profile, with the terminal ends 472a, 472b, 476a, 476b of the slots 472, 476 partially surrounding the ends 478a, 478b of terminal ends of the opening 478. In one aspect, slot 472 is a first TCR adjustment slot, slot 476 is a second TCR adjustment slot, and slot 478 is a third TCR adjustment slot.

[0163] The TCR adjustment slots 472, 476 are aligned along a longitudinal axis (similar to axis L3 of FIG. 4). Midpoints of each of the TCR adjustment slots 472, 476 can be centered on the longitudinal axis. The TCR adjustment slots 472, 476 can be continuously curved. In one aspect, the TCR adjustment slots 472, 476 are mirror images of each other. The TCR adjustment slots 472, 476 are shown having a continuous arc profile. In certain aspects, continuously curved or a continuous arc profile may be considered as a curved or arcing profile or shape that does not include any 45 degree or right-angle turns or bends from a first terminal end to a second terminal end. In one aspect, the TCR adjustment slots 472, 476 have a constant radius and extend for an arc length of greater than 90 degrees. The TCR adjustment slots 472, 476 are arranged in a medial area of the resistor relative to the lateral axis, in one aspect. The TCR adjustment slots 472, 476 are symmetrical with each other, in one aspect. In one aspect, each of the TCR adjustment slots 472, 476 are symmetrical about the longitudinal axis of the resistor. Each of the TCR adjustment slots 472, 476 may be intersected by the longitudinal axis at a respective midpoint.

[0164] All aspects of the slots 472, 476, 478 may vary. For example, the slots 472, 476, 478 may each have a different profile, dimension, size, or shape. These may be adjusted or changed based on need or function. The slots 472, 476, 478 may include one or more through openings and one or more openings that do not extend completely through the conductive portions and/or the resistive elements. The slots 472, 476, 478 may be circular, elongated, slotted, non-circular, S-shaped, N-shaped, serpentine, or any other shape as may be selected. The orientation or direction in which any of the slotted openings or elongated slots 472, 476, 478 extend may vary, having differences in height, length, width, etc. In certain aspects, one or more of the slots 472, 476, 478 may be omitted. These variations can be realized in any one or more of the other embodiments described herein.

[0165] As shown in FIG. 5, the slots 472, 476, and 478 are all arranged in an interior area of the resistor 410. The openings 472, 476, and 478 are each spaced away from outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor 410. No portion of the openings 472, 476, and 478 intersect, contact or cross the outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor 410. In this way, the openings 472, 476, and 478 are surrounded on all sides by parts of the conductive portions, the resistive elements, or combinations of those.

[0166] FIG. 19 illustrates a resistor 1910 that is similar to the resistor 410 of FIG. 5. Resistor 1910 is generally formed of a first resistive element 1911 disposed between a first conductive portion 1912 and a second conductive portion 1914, and a second resistive element 1913 disposed between the second conductive portion 1914 and a third conductive portion 1916. The resistor 1910 includes first, second, and third TCR adjustment slots 1972, 1976, 1978, as well as a first and second outer openings 1970, 1974. The configuration, structure, function, and other characteristics of all elements of resistor 1910 in FIG. 19 are similar to the corresponding elements of resistor 410 in FIG. 5.

[0167] As shown in FIG. 19, the openings 1970, 1972, 1974, 1976, and 1978 are all arranged in an

interior area of the resistor **1910**. The openings **1970**, **1972**, **1974**, **1976**, and **1978** are each spaced away from outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **1910**. No portion of the openings **1970**, **1972**, **1974**, **1976**, and **1978** intersect, contact or cross the outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **1910**. In this way, the openings **1970**, **1972**, **1974**, **1976**, and **1978** are surrounded on all sides by parts of the conductive portions, the resistive elements, or combinations of those.

[0168] FIG. **6** illustrates another embodiment of a resistor **510**. The resistor **510** is generally formed of a resistive element **511** disposed between a first conductive portion **512** and a second conductive portion **514**. The resistor **510** is similar to the resistor **410** except for the following modifications. The extent or length of the slots **572**, **576** are shorter than the extent of the slots **472**, **476**. The central slot **578** is oblong in shape and does not include enlarged ends as in the “dog bone” or barbell variation. Instead, the ends **578a**, **578b** are generally aligned and linear. The ends **572a**, **572b**, **576a**, **576b** are curved and partially surround the respective ends **578a**, **578b** of the central slot **578**. Slot **578** acts as a further TCR adjustment opening or slot. In one aspect, slot **572** is a first TCR adjustment slot, slot **576** is a second TCR adjustment slot, and slot **578** is a third TCR adjustment slot. In one aspect, the first and second TCR adjustment slots **572**, **576** are symmetrical about the longitudinal axis of the resistor. The first and second TCR adjustment slots **572**, **576** may be intersected by the longitudinal axis at a respective midpoint.

[0169] The TCR adjustment slots **572**, **576** are aligned along a longitudinal axis (similar to axis **L3** of FIG. **4**). Midpoints of each of the TCR adjustment slots **572**, **576** can be centered on the longitudinal axis. The TCR adjustment slots **572**, **576** can be continuously curved. In one aspect, the TCR adjustment slots **572**, **576** are mirror images of each other. The TCR adjustment slots **572**, **576** are shown having a continuous arc profile. In one aspect, the TCR adjustment slots **572**, **576** have a constant radius and extend for an arc length of greater than 90 degrees. The TCR adjustment slots **572**, **576** are arranged in a medial area of the resistor relative to the lateral axis, in one aspect. The TCR adjustment slots **572**, **576** are symmetrical with each other, in one aspect.

[0170] All aspects of the slots **572**, **576**, **578** may also vary. For example, the slots **572**, **576**, **578** may each have a different profile, dimension, size, or shape. These may be adjusted or changed based on need or function. The slots **572**, **576**, **578** may include one or more through openings and one or more openings that do not extend completely through the conductive portions and/or the resistive elements. The slots **572**, **576**, **578** may be circular, elongated, slotted, non-circular, S-shaped, N-shaped, serpentine, or any other shape as may be selected. The orientation or direction in which any of the slotted openings or elongated slots **572**, **576**, **578** extend may vary, having differences in height, length, width, etc. In certain aspects, one or more of the slots **572**, **576**, **578** may be omitted. These variations can be realized in any one or more of the other embodiments described herein.

[0171] As shown in FIG. **6**, the slots **572**, **576**, and **578** are all arranged in an interior area of the resistor **510**. The openings **572**, **576**, and **578** are each spaced away from outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **510**. No portion of the openings **572**, **576**, and **578** intersect, contact or cross the outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **510**. In this way, the openings **572**, **576**, and **578** are surrounded on all sides by parts of the conductive portions, the resistive elements, or combinations of those.

[0172] FIG. **20** illustrates a resistor **2010** that is similar to the resistor **510** of FIG. **6**. Resistor **2010** is generally formed of a first resistive element **2011** disposed between a first conductive portion **2012** and a second conductive portion **2014**, and a second resistive element **2013** disposed between the second conductive portion **2014** and a third conductive portion **2016**. The resistor **2010** includes first, second, and third TCR adjustment slots **2072**, **2076**, **2078**, as well as a first and second outer openings **2070**, **2074**. The configuration, structure, function, and other characteristics of all elements of resistor **2010** in FIG. **20** are similar to the corresponding elements of resistor **510** in FIG. **6**.

[0173] As shown in FIG. 20, the openings **2070**, **2072**, **2074**, **2076**, and **2078** are all arranged in an interior area of the resistor **2010**. The openings **2070**, **2072**, **2074**, **2076**, and **2078** are each spaced away from outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **2010**. No portion of the openings **2070**, **2072**, **2074**, **2076**, and **2078** intersect, contact or cross the outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **2010**. In this way, the openings **2070**, **2072**, **2074**, **2076**, and **2078** are surrounded on all sides by parts of the conductive portions, the resistive elements, or combinations of those.

[0174] FIG. 7 illustrates another embodiment of a resistor **610**. The resistor **610** is generally formed of a resistive element **611** disposed between a first conductive portion **612** and a second conductive portion **614**. A first and second C-shaped slot **672** and **676** are provided on the first and second conductive portions **612**, **614**, respectively. As shown in FIG. 7, an angle of curvature α of the curved ends **672a**, **672b**, **676a**, **676b** is angled relative to a medial or middle portion of the slots **672**, **676**. In one embodiment, the angle of curvature α is 65 degrees. The angle of curvature α can be 40 degrees to 90 degrees in another aspect. The overall angle defined terminal ends of each of the C-shaped slots **672**, **676** is greater than 90 degrees, in one aspect. The C-shaped slots **672** and **676** are concave towards a central region of the resistor **610**. This resistor **610** lacks any central opening or holes. All other aspects of the resistor **610** are similar to the aspects described herein with respect to the other embodiments of resistors. In one embodiment, pins are soldered next to the resistive element **611**, in an area between the slots **672**, **676** and the resistive element **611**. This arrangement could be implemented in all of the other embodiments.

[0175] The TCR adjustment slots **672**, **676** are aligned along a longitudinal axis (similar to axis L3 of FIG. 4). Midpoints of each of the TCR adjustment slots **672**, **676** can be centered on the longitudinal axis. The TCR adjustment slots **672**, **676** can be continuously curved. In one aspect, the TCR adjustment slots **672**, **676** are mirror images of each other. The TCR adjustment slots **672**, **676** are shown having a continuous arc profile. In one aspect, the TCR adjustment slots **672**, **676** have a constant radius and extend for an arc length of greater than 90 degrees. The TCR adjustment slots **672**, **676** are arranged in a medial area of the resistor relative to the lateral axis, in one aspect. The TCR adjustment slots **672**, **676** are symmetrical with each other, in one aspect. In one aspect, the TCR adjustment slots **672**, **676** are symmetrical about the longitudinal axis of the resistor. The TCR adjustment slots **672**, **676** may be intersected by the longitudinal axis at a respective midpoint.

[0176] All aspects of the slots **672**, **676** may also vary. For example, the slots **672**, **676** may each have a different profile, dimension, size, or shape. These may be adjusted or changed based on need or function. The slots **672**, **676** may include one or more through openings and one or more openings that do not extend completely through the conductive portions and/or the resistive elements. The slots **672**, **676** may be circular, elongated, slotted, non-circular, S-shaped, N-shaped, serpentine, or any other shape as may be selected. The orientation or direction in which any of the slotted openings or elongated slots **672**, **676** extend may vary, having differences in height, length, width, etc. In certain aspects, one or more of the slots **672**, **676** may be omitted. These variations can be realized in any one or more of the other embodiments described herein.

[0177] As shown in FIG. 7, the slots **672**, **676** are all arranged in an interior area of the resistor **610**. The openings **672**, **676** are each spaced away from outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **610**. No portion of the openings **672**, **676** intersect, contact or cross the outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **610**. In this way, the openings **672**, **676** are surrounded on all sides by parts of the conductive portions, the resistive elements, or combinations of those.

[0178] FIG. 8 illustrates another embodiment of a resistor **710**. The resistor **710** is generally formed of a resistive element **711** disposed between a first conductive portion **712** and a second conductive portion **714**. A first and second C-shaped slot **772** and **776** are provided on the first and second conductive portions **712**, **714**, respectively. As shown in FIG. 7, the angle of the curved ends **772a**, **772b**, **776a**, **776b** are angled relative to a medial or middle portion of the slots **772**, **776**. In one

embodiment, the angle of curvature **a** is 45 degrees. The angle of curvature **a** can be 20 degrees to 70 degrees in another aspect. A central slot **778** is generally centrally located through the resistor **710**, which extends from the first conductive portion **712** through the resistive element **711**, and the second conductive portion **714**. The central slot **778** may have a “dog bone” or barbell shape, having a larger portion **778a** at the left side (i.e. the X1 side), and smaller portion at its central or middle area **778b**, and a larger portion **778c** at the right side (i.e. the X2 side). The central slot **778** could also be rotated 90 degrees from the position shown in FIG. **8** in another variation. The C-shaped slots **772** and **776** are concave towards a central region of the resistor **710** and towards the central slot **778**. All other aspects of the resistor **710** are similar to the aspects described herein with respect to the other embodiments of resistors.

[0179] The TCR adjustment slots **772**, **776** are aligned along a longitudinal axis (similar to axis L3 of FIG. **4**). Midpoints of each of the TCR adjustment slots **772**, **776** can be centered on the longitudinal axis. The TCR adjustment slots **772**, **776** can be continuously curved. In one aspect, the TCR adjustment slots **772**, **776** are mirror images of each other. The TCR adjustment slots **772**, **776** are shown having a continuous arc profile. In one aspect, the TCR adjustment slots **772**, **776** have a constant radius and extend for an arc length of greater than 90 degrees. The TCR adjustment slots **772**, **776** are arranged in a medial area of the resistor relative to the lateral axis, in one aspect. The TCR adjustment slots **772**, **776** are symmetrical with each other, in one aspect. In one aspect, slot **772** is a first TCR adjustment slot, slot **776** is a second TCR adjustment slot, and slot **778** is a third TCR adjustment slot. In one aspect, the first and second TCR adjustment slots **772**, **776** are symmetrical about the longitudinal axis of the resistor. The first and second TCR adjustment slots **772**, **776** may be intersected by the longitudinal axis at a respective midpoint.

[0180] All aspects of the slots **772**, **776**, **778** may also vary. For example, the slots **772**, **776**, **778** may each have a different profile, dimension, size, or shape. These may be adjusted or changed based on need or function. The slots **772**, **776**, **778** may include one or more through openings and one or more openings that do not extend completely through the conductive portions and/or the resistive elements. The slots **772**, **776**, **778** may be circular, elongated, slotted, non-circular, S-shaped, N-shaped, serpentine, or any other shape as may be selected. The orientation or direction in which any of the slotted openings or elongated slots **772**, **776**, **778** extend may vary, having differences in height, length, width, etc. In certain aspects, one or more of the slots **772**, **776**, **778** may be omitted. These variations can be realized in any one or more of the other embodiments described herein.

[0181] As shown in FIG. **8**, the slots **772**, **776**, **778** are all arranged in an interior area of the resistor **710**. The openings **772**, **776**, **778** are each spaced away from outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **710**. No portion of the openings **772**, **776**, **778** intersect, contact or cross the outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **710**. In this way, the openings **772**, **776**, **778** are surrounded on all sides by parts of the conductive portions, the resistive elements, or combinations of those.

[0182] FIG. **9** illustrates another embodiment of a resistor **810**. The resistor **810** is generally formed of a resistive element **811** disposed between a first conductive portion **812** and a second conductive portion **814**. A first slot **872** is defined on the first conductive portion **812**. A central slot **878** is defined with a first end **878a** in the first conductive portion **812**, a middle portion **878b** defined in the resistive element **811**, and a second end **878c** in the second conductive portion **814**. A second slot **876** is defined on the second conductive portion **814**. As compared to FIG. **8**, the resistor **810** in FIG. **9** includes a straight or linear central slot **872**. The curved ends **872a**, **872b**, **876a**, **876b** of the slots **872**, **876** each partially surround respective ends of the central opening **878**. The C-shaped slots **872** and **876** are concave towards a central region of the resistor **810** and towards the central opening **878**. All other aspects of the resistor **810** are similar to the aspects described herein with respect to the other embodiments of resistors.

[0183] The TCR adjustment slots **872**, **876** are aligned along a longitudinal axis (similar to axis L3

of FIG. 4). Midpoints of each of the TCR adjustment slots **872**, **876** can be centered on the longitudinal axis. The TCR adjustment slots **872**, **876** can be continuously curved. In one aspect, the TCR adjustment slots **872**, **876** are mirror images of each other. The TCR adjustment slots **872**, **876** are shown having a continuous arc profile. In one aspect, the TCR adjustment slots **872**, **876** have a constant radius of curvature and extend for an arc length of greater than 90 degrees. The TCR adjustment slots **872**, **876** are arranged in a medial area of the resistor relative to the lateral axis, in one aspect. The TCR adjustment slots **872**, **876** are symmetrical with each other, in one aspect. In one aspect, slot **872** is a first TCR adjustment slot, slot **876** is a second TCR adjustment slot, and slot **878** is a third TCR adjustment slot. In one aspect, the first and second TCR adjustment slots **872**, **876** are symmetrical about the longitudinal axis of the resistor. The first and second TCR adjustment slots **872**, **876** may be intersected by the longitudinal axis at a respective midpoint.

[0184] All aspects of the slots **872**, **876**, **878** may also vary. For example, the slots **872**, **876**, **878** may each have a different profile, dimension, size, or shape. These may be adjusted or changed based on need or function. The slots **872**, **876**, **878** may include one or more through openings and one or more openings that do not extend completely through the conductive portions and/or the resistive elements. The slots **872**, **876**, **878** may be circular, elongated, slotted, non-circular, S-shaped, N-shaped, serpentine, or any other shape as may be selected. The orientation or direction in which any of the slotted openings or elongated slots **872**, **876**, **878** extend may vary, having differences in height, length, width, etc. In certain aspects, one or more of the slots **872**, **876**, **878** may be omitted. These variations can be realized in any one or more of the other embodiments described herein.

[0185] As shown in FIG. 9, the slots **872**, **876**, **878** are all arranged in an interior area of the resistor **810**. The openings **872**, **876**, **878** are each spaced away from outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **810**. No portion of the openings **872**, **876**, **878** intersect, contact or cross the outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **810**. In this way, the openings **872**, **876**, **878** are surrounded on all sides by parts of the conductive portions, the resistive elements, or combinations of those.

[0186] Upper and lower notches in the resistive elements, as best shown in FIGS. 5 and 6, may be used to adjust the base value of the resistor, which also will affect the TCR value. These features and the shadowing feature can be “tuned” or adjusted in conjunction with each other to achieve a desired value or result.

[0187] A method **1000** of making a resistor according to aspects of the invention having a first resistive element and a second resistive element is provided. The method **1000** comprises at least the following steps, as shown in the flowchart of FIG. 10.

[0188] At step **1010**, a first resistive element is disposed between a first conductive portion at one side, and a second conductive portion at an opposite second side.

[0189] At step **1020**, a second resistive element is disposed between the second conductive portion on one side, and a third conductive portion at an opposite second side.

[0190] At step **1030**, a first opening is created in the first conductive portion adjacent an outer portion of the first conductive portion, and a second opening is created adjacent an inner portion of the first conductive portion. The second opening in the first conductive portion is provided as a TCR adjustment opening or slot.

[0191] At step **1040**, an opening is created in the second conductive portion. The opening in the second conductive portion is provided as a TCR adjustment opening slot.

[0192] At step **1050**, a first opening is created in the third conductive portion adjacent an outer portion of the third conductive portion, and a second opening is created adjacent an inner portion of the third conductive portion. The second opening in the third conductive portion is provided as a TCR adjustment opening or slot.

[0193] At step **1060**, the TCR value of the resistor may be adjusted by modifying the size and/or

shape and/or position of the TCR adjustment openings or slots. The sizes, shapes and positions of the various TCR adjustment openings or slots may be as described previously herein.

[0194] At step **1070**, one or more slots may be formed in a portion of the first resistive element configured to adjust the resistance value of the resistor. A slot may be formed in a portion of the second resistive element configured to adjust the resistance value of the resistor.

[0195] A method **1100** of making a resistor according to aspects of the invention having a resistive element is provided. The method **1100** comprises at least the following steps, as shown in the flowchart of FIG. **11**.

[0196] At step **1110**, a resistive element is disposed between a first conductive portion at one side, and a second conductive portion at an opposite second side.

[0197] At step **1120**, a first opening is created in the first conductive portion adjacent an outer portion of the first conductive portion, and a second opening is created adjacent an inner portion of the first conductive portion. The second opening in the first conductive portion is provided as a TCR adjustment opening or slot.

[0198] At step **1130**, a first opening is created in the second conductive portion adjacent an outer portion of the second conductive portion, and a second opening is created adjacent an inner portion of the third conductive portion. The second opening in the third conductive portion is provided as a TCR adjustment opening or slot.

[0199] At step **1140**, an opening is created in the resistive element. The opening is provided as a TCR adjustment opening or slot. The opening may be provided in the resistive element and may also extend through the first conductive portion and the second conductive portion.

[0200] At step **1150**, the TCR value of the resistor may be adjusted by modifying the size and/or shape and/or position of the TCR adjustment openings or slots. The sizes, shapes and positions of the various TCR adjustment openings or slots may be as described previously herein.

[0201] At step **1160**, one or more slots may be formed in a portion of the first resistive element configured to adjust the resistance value of the resistor.

[0202] FIG. **27** illustrates a method **2700** of forming a resistor according to another aspect. As shown in FIG. **27**, step **2710** includes providing a first resistive element disposed between a first conductive portion and a second conductive portion. Step **2720** includes forming a first TCR opening in the first conductive portion and forming a second TCR opening in the second conductive portion. In one aspect, the first TCR adjustment opening and the second TCR adjustment opening are each formed as an elongated slot formed with a continuous arc profile.

[0203] Step **2730** includes adjusting at least one characteristic of at least one of the first TCR opening or the second TCR opening to provide for a final TCR value, or an adjusted TCR value, of the resistor. This step can include optimization processes that are configured to adjust and configure the TCR value of the resistor. In one aspect, optimizing the TCR value of the resistor can be achieved through various processes and methods, such as experimentation, trials, and prototyping, which may include various types of monitoring, sensing and feedback.

[0204] Step **2740** includes forming a third TCR adjustment opening. In one aspect, the third TCR adjustment opening is partially defined on each of the first resistive element, the first conductive portion, and the second conductive portion. The third TCR adjustment opening is configured to adjust the TCR value of the resistor.

[0205] FIGS. **13A-13E** illustrate various embodiments of resistors **1310a-1310e**. In FIGS. **13A**, **13B**, and **13E**, first conductive portions **1312a**, **1312b**, **1312e**, second conductive portions **1314a**, **1314b**, **1324e** are provided with a resistive element **1311a**, **1311b**, **1311c** therebetween. In FIGS. **13C** and **13D**, first conductive portions **1312c**, **1312d**, second conductive portions **1314c**, **1314d**, and third conductive portions **1316c**, **1316d** are provided and two resistive elements **1311c**, **1313c**, **1311d**, **1313d** are sandwiched between respective ones of the conductive portions. In FIGS. **13A-13E**, a first plurality of voltage sense points **1390a**, second plurality of voltage sense points **1390b**, third plurality of voltage sense points **1390c**, fourth plurality of voltage sense points **1390d**, and

fifth plurality of voltage sense points **1390e** are provided in respective areas of the openings or slots **1378a**, **1378b**, **1372c**, **1378c**, **1376c**, **1372d**, **1378d**, **1376d**, **1372e**, **1376e**. One of ordinary skill in the art would appreciate and understand that other openings or slots of the resistors can include voltage sense points. In one aspect, the voltage sense points are configured to define regions or locations in which voltage sensing elements, such as sensing pins, can be arranged. The voltage sense points can be defined via openings or holders defined in brackets, bracket assemblies, headers, or header assemblies. Additionally, the left and right sides of the resistors **1310a-1310e** provide current connections.

[0206] As shown in FIGS. **13A-13E**, the various openings are all arranged in an interior area of the resistor **1310a-1310e**. The openings are each spaced away from outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **1310a-1310e**. No portion of the openings intersect, contact or cross the outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **1310a-1310e**. In this way, the openings are surrounded on all sides by parts of the conductive portions, the resistive elements, or combinations of those.

[0207] FIG. **14** is a chart illustrating measurements for a redundant TCR adjustable shunt with optimized voltage sense isolation having two resistive elements. As shown in FIG. **14**, Elements **1** and **2** generally have a 0% or nearly 0% change in a range of temperatures from -40 degrees Celsius to 125 degrees Celsius.

[0208] FIG. **15** is a chart illustrating measurements for a shunt with different amounts or levels of voltage sense isolation. FIG. **15** illustrates three different sets of measurements, including situations with: no isolation for sense points; small isolation for sense points; and large isolation for sense points. As shown in FIG. **15**, each of the three sets of data essentially converge to a 0% change at roughly 25 degrees Celsius. The percentage change generally stays negative for the situation with large isolation for sense points at negative temperatures and high temperatures. In contrast, in the situation with no isolation for sense points, the percentage change starts below -1.00% around -55 degrees Celsius and increases to above 0.75% at extreme high temperatures, i.e. roughly 135 degrees Celsius. In the situation with small isolation for sense points, at roughly -55 degrees Celsius, the percentage change is roughly -0.75% and increases to roughly 0.40% at extreme high temperatures, i.e. roughly 135 degrees Celsius.

[0209] In another aspect of the invention, a shunt resistor may be configured in a generally cylindrical arrangement. As shown in FIG. **16**, such a cylindrical shunt **1600** comprises a first generally tubular conductive portion **1610**, a first resistive element **1611** that is generally cylindrical in shape, a second or central generally tubular conductive portion **1612**, a second resistive element **1613** that is generally cylindrical in shape, and a third generally tubular conductive portion **1614**. The shunt resistor of FIG. **16** improves upon the arrangements shown in U.S. Pat. No. 8,344,846, the entire contents of which is incorporated by reference herein.

[0210] The first generally tubular conductive portion **1610** has a first open end **1616** facing in the first direction **X1**, and a second open end **1618** facing in the second direction **X2** having an opening configured to receive the first resistive element **1611**. A current sense lead **1620** may extend from or otherwise be attached or connected to an outer surface of a wall of the first generally tubular conductive portion **1610** adjacent the second end. The first open end **1616** may be configured to attach to an external connection, such as a cable.

[0211] The first resistive element **1611** may be formed as a cylindrical wire formed from a resistive material as described herein. The length and other dimensions of the first resistive element **1611** may be selected based on need, function or performance. The first resistive element **1611** is connected at a first end **1622** to the second open end **1618** of the first generally tubular conductive portion **1610**, which is configured to receive and be connected to the first end **1622** of the first resistive element **1611**. The connection may be by, for example, welding, such as magnetic pulse welding.

[0212] The second generally tubular conductive portion **1612** has a first open end **1624** facing in

the first direction X1, and a second open end **1626** facing in the second direction X2, and may be essentially a ring shape. A current sense lead **1628** may extend from or otherwise be attached or connected to an outer surface of a wall of the second generally tubular conductive portion **1612**. The first open end **1624** is configured to receive and connect to a second end **1630** of the first resistive element **1611**. The connection may be by, for example welding, such as magnetic pulse welding.

[0213] The second resistive element **1613** may be formed as a cylindrical wire formed from a resistive material as described herein. The length and other dimensions of the second resistive element **1613** is connected at a first end **1632** to the second open end **1626** of the second generally tubular conductive portion **1612**. The connection may be by, for example welding, such as magnetic pulse welding.

[0214] The third generally tubular conductive portion **1614** has a first open end **1634** facing in the first direction X2, and a second open end **1636** facing in the second direction X1 and having an opening configured to receive and to be connected to a second end **1638** of the second resistive element **1613**. A current sense lead **1640** may extend from or otherwise be attached or connected to an outer surface of a wall of the third generally tubular conductive portion **1614** adjacent the second end **1636**. The first open end **1634** may be configured to attach to an external connection, such as a cable.

[0215] Regarding the cylindrical shunt aspect, TCR compensation may be provided based on reduced distances due to various elements of the design. For example, adjustments of the thickness of the device provide in part for TCR compensation. TCR compensation may also be provided due to placement of the voltage connections or leads, or sense connections or leads, with respect to transitional edges between conductive portions and the multiple resistive elements. The inclusion of multiple resistive elements, i.e. the redundant configuration, may also improve TCR compensation.

[0216] Adjustment of the resistance value can be made by a lathe, abrasion, or lasering operation that reduces the diameter of the exposed resistant element wires or by removing conductive material such as copper from the end of each tubular portion effectively lengthening the resistive element. In addition, resistance can be adjusted by adding termination material (such as copper or other conductive material) back to the resistive element such as by welding. Slots or openings may also be provided at various locations, to allow for additional TCR adjustments and fine tuning.

[0217] FIGS. **17A-17M** illustrate multiple variations of resistors that function the same as the other embodiments described herein and are provided to show various configurations.

[0218] Each of the resistors **1710a-1710m** in FIGS. **17A-17M** includes a first conductive portion **1712a-1712m** (i.e. the left-most conductive portion), a second conductive portion **1714a-1714m** (i.e. the middle or medial conductive portion), and a third conductive portion **1716a-1716m** (i.e. the right-most conductive portion). A first resistive element **1711a-1711m** is arranged between the first conductive portion **1712a-1712m** and the second conductive portion **1714a-1714m**, and a second resistive element **1713a-1713m** is arranged between the second conductive portion **1714a-1714m** and the third conductive portion **1716a-1716m**. Each of the resistors **1710a-1710m** include TCR adjustment openings or slots, or TCR compensation openings or slots, which may include a first TCR adjustment opening or slot **1772a-1772m**, a second TCR adjustment opening or slot **1776a-1776m**, and a third TCR adjustment opening or slot **1778a-1778m**.

[0219] FIG. **17A** illustrates an aspect including a first, second and third conductive portion **1712a**, **1714a**, **1716a** and a first and a second resistive element **1711a**, **1713a**. In this embodiment, the left-most conductive portion **1712a** includes a current connection opening **1770a** and the right-most conductive portion **1716a** lacks a current connection opening. All other aspects of this embodiment are otherwise identical to the embodiment disclosed in FIG. **1**.

[0220] FIG. **17B** illustrates an aspect including a first, second and third conductive portion **1712b**,

1714b, **1716b** and a first and a second resistive element **1711b**, **1713b**. In this embodiment, the left-most conductive portion **1712b** lacks a current connection opening and the right-most conductive portion **1713** includes a current connection opening **1774b**. All other aspects of this embodiment are otherwise identical to the embodiment disclosed in FIG. 1.

[0221] FIG. 17C illustrates an aspect including a first, second and third conductive portion **1712c**, **1714c**, **1716c** and a first and a second resistive element **1711c**, **1713c**. In this embodiment, the left-most conductive portion **1712c** and the right-most conductive portion **1716c** both lack current connection openings. All other aspects of this embodiment are otherwise identical to the embodiment disclosed in FIG. 1.

[0222] FIG. 17D illustrates an aspect including a first, second and third conductive portion **1712d**, **1714d**, **1716d** and a first and a second resistive element **1711d**, **1713d**. In this embodiment, the left-most conductive portion **1712d** includes a first current connection opening **1770d** and the right-most conductive portion **1716d** includes a second current connection opening **1774d**. The second current connection opening **1774d** is larger than the first current connection opening **1770d**. In one aspect, the second current connection opening **1774d** is at least 100% larger than the first current connection opening **1770d**. All other aspects of this embodiment are otherwise identical to the embodiment disclosed in FIG. 1.

[0223] FIG. 17E illustrates an aspect including a first, second and third conductive portion **1712e**, **1714e**, **1716e** and a first and a second resistive element **1711e**, **1713e**. In this embodiment, the left-most conductive portion **1712e** includes a first current connection opening **1770e** and the right-most conductive portion **1716e** includes a second current connection opening **1774e**. The second current connection opening **1774e** is smaller than the first current connection opening **1770e**. In one aspect, the second current connection opening **1774e** is at least half the size of the first current connection opening **1770e**. All other aspects of this embodiment are otherwise identical to the embodiment disclosed in FIG. 1.

[0224] FIG. 17F illustrates an aspect including a first, second and third conductive portion **1712f**, **1714f**, **1716f** and a first and a second resistive element **1711f**, **1713f**. In this embodiment, the left-most conductive portion **1712f** includes a first current connection opening **1770f** and the right-most conductive portion **1716f** includes a second current connection opening **1774f**. The first current connection opening **1770f** has an oblong or elongated circular shape (i.e. a pill shape or rectangular ellipse), such as a slotted configuration, and the second current connection opening **1774f** has a perfect circular or round shape. All other aspects of this embodiment are otherwise identical to the embodiment disclosed in FIG. 1.

[0225] FIG. 17G illustrates an aspect including a first, second and third conductive portion **1712g**, **1714g**, **1716g** and a first and a second resistive element **1711g**, **1713g**. In this embodiment, the left-most conductive portion **1712g** includes a first current connection opening **1770g** and the right-most conductive portion **1716g** includes a second current connection opening **1774g**. The second current connection opening **1774g** has an oblong or elongated circular shape (i.e. a pill shape or rectangular ellipse), such as a slotted configuration, and the first current connection opening **1770g** has a perfect circular or round shape. All other aspects of this embodiment are otherwise identical to the embodiment disclosed in FIG. 1.

[0226] FIG. 17H illustrates an aspect including a first, second and third conductive portion **1712h**, **1714h**, **1716h** and a first and a second resistive element **1711h**, **1713h**. In this embodiment, the left-most conductive portion **1712h** includes a first current connection opening **1770h** and the right-most conductive portion **1716h** includes a second current connection opening **1774h**. The first current connection opening **1770h** and the second current connection opening **1774h** are identical and each have an oblong or elongated circular shape (i.e. a pill shape or rectangular ellipse), such as a slotted configuration. All other aspects of this embodiment are otherwise identical to the embodiment disclosed in FIG. 1.

[0227] FIG. 17I illustrates an aspect including a first, second and third conductive portion **1712i**,

1714i, **1716i** and a first and a second resistive element **1711i**, **1713i**. In this embodiment, the left-most conductive portion **1712i** includes a first current connection opening **1770i** and the right-most conductive portion **1716i** includes a second current connection opening **1774i**. The first current connection opening **1770i** has an elliptical shape and the second current connection opening **1774i** has a perfect circular or round shape. All other aspects of this embodiment are otherwise identical to the embodiment disclosed in FIG. 1.

[0228] FIG. 17J illustrates an aspect including a first, second and third conductive portion **1712j**, **1714j**, **1716j** and a first and a second resistive element **1711j**, **1713j**. In this embodiment, the left-most conductive portion **1712j** includes a first current connection opening **1770j** and the right-most conductive portion **1716j** includes a second current connection opening **1774j**. The second current connection opening **1774j** has an elliptical shape and the first current connection opening **1770j** has a perfect circular or round shape. All other aspects of this embodiment are otherwise identical to the embodiment disclosed in FIG. 1.

[0229] FIG. 17K illustrates an aspect including a first, second and third conductive portion **1712k**, **1714k**, **1716k** and a first and a second resistive element **1711k**, **1713k**. In this embodiment, the left-most conductive portion **1712k** includes a first current connection opening **1770k** and the right-most conductive portion **1716k** includes a second current connection opening **1774k**. The first current connection opening **1770k** and the second current connection opening **1774k** are identical and each have an elliptical shape. All other aspects of this embodiment are otherwise identical to the embodiment disclosed in FIG. 1.

[0230] FIG. 17L illustrates an aspect including a first, second and third conductive portion **1712l**, **1714l**, **1716l** and a first and a second resistive element **1711l**, **1713l**. In this embodiment, the left-most conductive portion **1712l** lacks any current connection opening and the right-most conductive portion **1716l** includes a current connection opening **1774l**. The current connection opening **1774l** of the right-most conductive portion **1716l** has an elliptical shape. All other aspects of this embodiment are otherwise identical to the embodiment disclosed in FIG. 1.

[0231] FIG. 17M illustrates an aspect including a first, second and third conductive portion **1712m**, **1714m**, **1716m** and a first and a second resistive element **1711m**, **1713m**. In this embodiment, the right-most conductive portion **1716m** lacks any current connection opening and the left-most conductive portion **1712m** includes a current connection opening **1770m**. The current connection opening **1770m** of the left-most conductive portion **1712m** has an elliptical shape. All other aspects of this embodiment are otherwise identical to the embodiment disclosed in FIG. 1.

[0232] As shown in FIGS. 17A-17M, the various openings are all arranged in an interior area of the resistor **1710a-1710m**. The openings are each spaced away from outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **1710a-1710m**. No portion of the openings intersect, contact or cross the outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **1710a-1710m**. In this way, the openings are surrounded on all sides by parts of the conductive portions, the resistive elements, or combinations of those.

[0233] FIGS. 21A-24B generally illustrate configurations in which sensing pins are used in conjunction with resistors. These configurations may be used, in one aspect, in order to provide sensing pins within a certain voltage range of an analog to digital converter (ADC) of the ADC's analog ground reference input. In one aspect, a sense circuit may have high impedance inputs to limit current flow. The current flow within a sense circuit causes voltage drops that will affect the voltage reading of the ADC. One aspect achieving an optimized circuit configuration is through the use of an additional pin connected to the ground side of the battery shunt, that is arranged outside of the main sensing circuit. In one aspect, this design allows for a low voltage differential between the ADC analog ground reference and the sense pin ground, and thereby the current flow through the sense circuit.

[0234] For example, if the components for the sensing device use 10 mA of current when powered, then 10 mA is passed through the ground pin. If there is a minimal 1 mΩ of resistance in between

the PCB pin connection and the shunt itself, it creates a 10 μ V drop. When sensing a 100 $\mu\Omega$ shunt, a 10 μ V drop equates to a 100 mA reading error. This error will vary part to part based on the PCB to shunt resistance and will vary within a single part over time based on the device power drawn under different operating modes. One way to address sources of error is through the use of a third or additional ground pin on the shunt, which is shown, in part, by the embodiments of FIGS. **21A**, **21B**, and **22**.

[0235] FIGS. **21A** and **21B** illustrate another aspect of a resistor **2110**. The resistor **2110** is generally formed of a resistive element **2111** disposed between a first conductive portion **2112** and a second conductive portion **2114**. The resistive element **2111** may be welded or otherwise bonded between the first conductive portion **2112** and the second conductive portion **2114**. The resistor **2110** may be similar to the resistor **110** of FIG. **2** and has a similar function and structure to resistor **110** unless otherwise specified.

[0236] As shown in FIGS. **21A** and **21B**, a first TCR adjustment opening **2172** and a second TCR adjustment opening **2176** are formed on the first conductive portion **2112** and the second conductive portion **2114** respectively, and current connections or connection areas are provided via a first outer circular hole **2170** on the first conductive portion **2112** and a second outer circular opening **2174** on the second conductive portion **2114**. FIGS. **21A** and **21B** illustrate a single current sense resistor, that is, there is a single resistive element.

[0237] All aspects of the openings **2170**, **2172**, **2174**, **2176** may vary. For example, the openings **2170**, **2172**, **2174**, **2176** may each have a different profile, dimension, size, or shape. These may be adjusted or changed based on need or function. The openings **2170**, **2172**, **2174**, **2176** may include one or more through openings and one or more openings that do not extend completely through the conductive portions and/or the resistive elements. The openings **2170**, **2172**, **2174**, **2176** may be circular, elongated, slotted, non-circular, S-shaped, N-shaped, serpentine, or any other shape as may be selected. The orientation or direction in which any of the slotted or elongated openings **2170**, **2172**, **2174**, **2176** extend may vary, having differences in height, length, width, etc. In certain aspects, one or more of the openings **2170**, **2172**, **2174**, **2176** may be omitted. These variations can be realized in any one or more of the other embodiments described herein.

[0238] The openings **2170**, **2172**, **2174**, **2176** are all arranged in an interior area of the resistor **2110**. The openings **2170**, **2172**, **2174**, **2176** are each spaced away from outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **2110**. No portion of the openings **2170**, **2172**, **2174**, **2176** intersect, contact or cross the outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **2110**. In this way, the openings **2170**, **2172**, **2174**, **2176** are surrounded on all sides by parts of the conductive portions, the resistive elements, or combinations of those.

[0239] As shown in FIGS. **21A** and **21B**, a header or header assembly **2180** is disclosed that is configured to provide a convenient arrangement for securing pins, such as a first pin **2184a**, a second pin **2184b**, and a third pin **2184c**. The header assembly **2180** includes braces having a plurality of openings, i.e. a first opening **2181a**, second opening **2181b**, third opening **2181c**, fourth opening **2181d**, and fifth opening **2181e**, configured to attach to or otherwise secure at least one of the pins **2184a**, **2184b**, **2184c**. The openings **2181a**-**2181e** define predetermined pin location or pin placement locations. As shown in FIGS. **21A** and **21B** only the first, fourth, and fifth openings **2181a**, **2181d**, **2181e** are occupied by a respective one of the pins **2184a**, **2184b**, **2184c**. This configuration is advantageous for manufacturing and allows soldering of the pins **2184a**, **2184b**, **2184c** to inner or inside surfaces of openings **2172**, **2176**, which further improve TCR performance. This arrangement also assists with positioning of the pins **2184a**, **2184b**, **2184c** relative to the resistor **2110**, and further provides an insulative spacing part in the header assembly **2180**.

[0240] As shown in FIGS. **21A** and **21B**, first and second pins **2184a** and **2184b** are configured as sensing pins and third pin **2184c** is configured as a ground pin. Cross-sectional hatching is shown

in FIG. 21B for ground pin **2184c** only for illustrative purposes in order to differentiate the ground pin **2184c** from the sensing pins **2184a** and **2184b**. The sensing pins **2184a**, **2184b** are preferably arranged so as to contact inner surface edges of the openings **2172**, **2176** that are closest to each other, as shown in FIGS. 21A and 21B. Ground pin **2184c** is arranged so as to contact an outermost surface edge of opening **2176**. The ground pin **2184c** generally provides a ground reference for a measurement system that is separate from the rest of the resistor. The sensing pins **2184a**, **2184b** are configured to be connected, such as via soldering, to a printed circuit board (PCB) or connector that is attached to the PCB. In one aspect, the sensing pins **2184a**, **2184b** are configured as voltage sense pins and are configured to be connected to a positive and negative voltage section of a sensing circuit.

[0241] Although three pins **2184a**, **2184b**, **2184c** are illustrated in FIGS. 21A and 21B and the pins **2184a**, **2184b**, **2184c** are shown in specific configurations, one of ordinary skill in the art would understand that the quantity, location, shape, orientation, and other aspects of the pins **2184a**, **2184b**, **2184c** can vary.

[0242] FIG. 22 illustrates a resistor **2210** having similar characteristics as the resistor **2110** of FIGS. 21A and 21B. As shown in FIG. 22, resistor **2210** is generally formed of a resistive element **2211** disposed between a first conductive portion **2212** and a second conductive portion **2214**. The resistive element **2211** may be welded or otherwise bonded between the first conductive portion **2212** and the second conductive portion **2214**. The resistor **2210** may be similar to the resistor **110** of FIG. 2 and has a similar same function and structure to resistor **110** unless otherwise specified.

[0243] As shown in FIG. 22, a first TCR adjustment opening **2272** and a second TCR adjustment opening **2276** are formed on the first conductive portion **2212** and the second conductive portion **2214**, respectively, and current connections or connection areas are provided via a first outer circular opening **2270** on the first conductive portion **2212** and a second outer circular opening **2274** on the second conductive portion **2214**. FIG. 22 illustrates a single current sense resistor, that is, there is a single resistive element.

[0244] All aspects of the openings **2270**, **2272**, **2274**, **2276** may vary. For example, the openings **2270**, **2272**, **2274**, **2276** may each have a different profile, dimension, size, or shape. These may be adjusted or changed based on need or function. The openings **2270**, **2272**, **2274**, **2276** may include one or more through openings and one or more openings that do not extend completely through the conductive portions and/or the resistive elements. The openings **2270**, **2272**, **2274**, **2276** may be circular, elongated, slotted, non-circular, S-shaped, N-shaped, serpentine, or any other shape as may be selected. The orientation or direction in which any of the slotted or elongated openings **2270**, **2272**, **2274**, **2276** extend may vary, having differences in height, length, width, etc. In certain aspects, one or more of the openings **2270**, **2272**, **2274**, **2276** may be omitted. These variations can be realized in any one or more of the other embodiments described herein.

[0245] The openings **2270**, **2272**, **2274**, **2276** are all arranged in an interior area of the resistor **2210**. The openings **2270**, **2272**, **2274**, **2276** are each spaced away from outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **2210**. No portion of the openings **2270**, **2272**, **2274**, **2276** intersect, contact or cross the outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **2210**. In this way, the openings **2270**, **2272**, **2274**, **2276** are surrounded on all sides by parts of the conductive portions, the resistive elements, or combinations of those.

[0246] As shown in FIG. 22, a header or header assembly **2280** is provided that is configured to provide a convenient arrangement for securing pins, such as a first pin **2284a**, a second pin **2284b**, and a third pin **2284c**. The header assembly **2280** includes braces having a plurality of openings, i.e. a first opening **2281a**, second opening **2281b**, third opening **2281c**, fourth opening **2281d**, and fifth opening **2281e**, configured to attach to or otherwise secure at least one of the pins **2284a**, **2284b**, **2284c**. As shown in FIG. 22, only the first, fourth, and fifth openings **2281a**, **2281d**, **2281e** are occupied by a respective one of the pins **2284a**, **2284b**, **2284c**. This configuration is

advantageous for manufacturing and allows soldering of the pins **2284a**, **2284b**, **2284c** to inner or inside surfaces of openings **2272**, **2276**, which further improve TCR performance.

[0247] As shown in FIG. **22**, the first and second pins **2284a**, **2284b** are configured as sensing pins and the third pin **2284c** is configured as a ground pin. The sensing pins **2284a**, **2284b** are preferably arranged so as to contact inner surface edges of the openings **2272**, **2276** that are closest to each other, as shown in FIG. **22**. Ground pin **2284c** is arranged so as to contact an outermost surface edge of opening **2276**. The ground pin **2284c** generally provides a ground reference for a measurement system that is separate from the rest of the resistor. The sensing pins **2284a**, **2284b** are configured to be connected, such as via soldering, to a printed circuit board (PCB) or connector that is attached to the PCB. In one aspect, the sensing pins **2284a**, **2284b** are configured as voltage sense pins and are configured to be connected to a positive and negative voltage section of a sensing circuit.

[0248] Although three pins **2284a**, **2284b**, **2284c** are illustrated in FIG. **22** and the pins **2284a**, **2284b**, **2284c** are shown in specific configurations, one of ordinary skill in the art would understand that the quantity, location, shape, orientation, and other aspects of the pins **2284a**, **2284b**, **2284c** can vary.

[0249] FIG. **23** illustrates a resistor **2310** having similar characteristics as the resistor **2110** of FIGS. **21A** and **21B**. As shown in FIG. **23**, resistor **2310** is generally formed of a resistive element **2311** disposed between a first conductive portion **2312** and a second conductive portion **2314**. The resistive element **2311** may be welded or otherwise bonded between the first conductive portion **2312** and the second conductive portion **2314**. The resistor **2310** may be similar to the resistor **110** of FIG. **2** and has a similar function and structure to resistor **110** unless otherwise specified.

[0250] As shown in FIG. **23**, a first TCR adjustment opening **2372** and a second TCR adjustment opening **2376** are formed on the first conductive portion **2312** and the second conductive portion **2314**, respectively, and current connections or connection areas are provided via outer circular openings **2370**, **2374** on the first conductive portion **2312** and the second conductive portion **2314**. As shown in FIG. **23**, the TCR adjustment opening **2372** has a quasi-slotted profile as compared to the TCR adjustment opening **2376**.

[0251] All aspects of the openings **2370**, **2372**, **2374**, **2376** may vary. For example, the openings **2370**, **2372**, **2374**, **2376** may each have a different profile, dimension, size, or shape. These may be adjusted or changed based on need or function. The openings **2370**, **2372**, **2374**, **2376** may include one or more through openings and one or more openings that do not extend completely through the conductive portions and/or the resistive elements. The openings **2370**, **2372**, **2374**, **2376** may be circular, elongated, slotted, non-circular, S-shaped, N-shaped, serpentine, or any other shape as may be selected. The orientation or direction in which any of the slotted or elongated openings **2370**, **2372**, **2374**, **2376** extend may vary, having differences in height, length, width, etc. In certain aspects, one or more of the openings **2370**, **2372**, **2374**, **2376** may be omitted. These variations can be realized in any one or more of the other embodiments described herein.

[0252] The openings **2370**, **2372**, **2374**, **2376** are all arranged in an interior area of the resistor **2310**. The openings **2370**, **2372**, **2374**, **2376** are each spaced away from outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **2310**. No portion of the openings **2370**, **2372**, **2374**, **2376** intersect, contact or cross the outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **2310**.

[0253] As shown in FIG. **23**, a first header assembly **2380a** and a second header assembly **2380b** are provided that are configured to provide a convenient arrangement for securing pins, such as a first pin **2384a**, second pin **2384b**, third pin **2384c**, fourth pin **2384d**, and fifth pin **2384e**. The header assemblies **2380a**, **2380b** includes braces having a plurality of openings, i.e. a first opening **2381a**, second opening **2381b**, third opening **2381c**, fourth opening **2381d**, fifth opening **2381e**, sixth opening **2381f**, seventh opening **2381g**, eighth opening **2381h**, ninth opening **2381i**, tenth opening **2381j**, eleventh opening **2381k**, and twelfth opening **2381l**, configured to attach to or

otherwise secure at least one of the pins **2384a**, **2384b**, **2384c**, **2384d**, **2384e**. As shown in FIG. 23, only the first opening **2381a**, second opening **2381b**, sixth opening **2381f**, eighth opening **2381h**, and twelfth opening **2381i** are occupied by a respective one of the pins **2384a**, **2384b**, **2384c**, **2384d**, **2384e**. This configuration is advantageous for manufacturing and allows soldering of the pins **2384a**, **2384b**, **2384c**, **2384d**, **2384e** to inner or inside surfaces of openings **2372**, **2376**, which further improve TCR performance. FIG. 23 illustrates a dual current sense resistor, which is considered a redundant voltage connection arrangement in one aspect.

[0254] As shown in FIG. 23, the second, third, fourth, and fifth pins **2384b**, **2384c**, **2384d**, **2384e** are configured as sensing pins and the first pin **2384a** is configured as a ground pin. The sensing pins **2384b**, **2384c**, **2384d**, **2384e** are preferably arranged so as to contact inner surface edges of the openings **2372**, **2376** that are closest to each other, as shown in FIG. 23. Ground pin **2384a** is arranged so as to contact an outermost surface edge of opening **2376**. The ground pin **2384a** generally provides a ground reference for a measurement system that is separate from the rest of the resistor. The sensing pins **2384b**, **2384c**, **2384d**, **2384e** are configured to be connected, such as via soldering, to a printed circuit board (PCB) or connector that is attached to the PCB. In one aspect, the sensing pins **2384b**, **2384c**, **2384d**, **2384e** are configured as voltage sense pins and are configured to be connected to a positive and negative voltage section of a sensing circuit.

[0255] Although five pins **2384a**, **2384b**, **2384c**, **2384d**, **2384e** are illustrated in FIG. 23 and the pins **2384a**, **2384b**, **2384c**, **2384d**, **2384e** are shown in specific configurations, one of ordinary skill in the art would understand that the quantity, location, shape, orientation, and other aspects of the pins **2384a**, **2384b**, **2384c**, **2384d**, **2384e** can vary.

[0256] FIGS. 24A and 24B illustrates a resistor **2410** having similar characteristics as the resistor **2110** of FIGS. 21A and 21B. As shown in FIGS. 24A and 24B, resistor **2410** is generally formed of a resistive element **2411** disposed between a first conductive portion **2412** and a second conductive portion **2414**. The resistive element **2411** may be welded or otherwise bonded between the first conductive portion **2412** and the second conductive portion **2414**. The resistor **2410** may be similar to the resistor **110** of FIG. 2 and has a similar function and structure to resistor **110** unless otherwise specified.

[0257] As shown in FIGS. 24A and 24B, a first TCR adjustment slot **2472** and a second TCR adjustment slot **2476** are formed on the first conductive portion **2412** and the second conductive portion **2414**, respectively, and current sense connections or connection areas are provided via outer circular openings **2470**, **2474** on the first conductive portion **2412** and the second conductive portion **2414**.

[0258] All aspects of the openings **2470**, **2472**, **2474**, **2476** may vary. For example, the openings **2470**, **2472**, **2474**, **2476** may each have a different profile, dimension, size, or shape. These may be adjusted or changed based on need or function. The openings **2470**, **2472**, **2474**, **2476** may include one or more through openings and one or more openings that do not extend completely through the conductive portions and/or the resistive elements. The openings **2470**, **2472**, **2474**, **2476** may be circular, elongated, slotted, non-circular, S-shaped, N-shaped, serpentine, or any other shape as may be selected. The orientation or direction in which any of the slotted or elongated openings **2470**, **2472**, **2474**, **2476** extend may vary, having differences in height, length, width, etc. In certain aspects, one or more of the openings **2470**, **2472**, **2474**, **2476** may be omitted. These variations can be realized in any one or more of the other embodiments described herein.

[0259] The openings **2470**, **2472**, **2474**, **2476** are all arranged in an interior area of the resistor **2410**. The openings **2470**, **2472**, **2474**, **2476** are each spaced away from outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **2410**. No portion of the openings **2470**, **2472**, **2474**, **2476** intersect, contact or cross the outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **2410**. In this way, the openings **2470**, **2472**, **2474**, **2476** are surrounded on all sides by parts of the conductive portions, the resistive elements, or combinations of those.

[0260] FIGS. 26A-26C illustrate a resistor **2610** having similar characteristics as the resistor **2110** of FIGS. 21A and 21B. As shown in FIGS. 26A-26C, the resistor **2610** is generally formed of a resistive element **2611** disposed between a first conductive portion **2612** and a second conductive portion **2614**. The resistive element **2611** may be welded or otherwise bonded between the first conductive portion **2612** and the second conductive portion **2614**. The resistor **2610** may be similar to the resistor **110** of FIG. 2 and has a similar function and structure to resistor **110** unless otherwise specified.

[0261] As shown in FIGS. 26A-26C, a first TCR adjustment opening **2672** and a second TCR adjustment opening **2676** are formed on the first conductive portion **2612** and the second conductive portion **2614**, respectively, and current connections or connection areas are provided via outer circular openings **2670**, **2674** on the first conductive portion **2612** and the second conductive portion **2614**.

[0262] All aspects of the openings **2670**, **2672**, **2674**, **2676** may vary. For example, the openings **2670**, **2672**, **2674**, **2676** may each have a different profile, dimension, size, or shape. These may be adjusted or changed based on need or function. The openings **2670**, **2672**, **2674**, **2676** may include one or more through openings and one or more openings that do not extend completely through the conductive portions and/or the resistive elements. The openings **2670**, **2672**, **2674**, **2676** may be circular, elongated, slotted, non-circular, S-shaped, N-shaped, serpentine, or any other shape as may be selected. The orientation or direction in which any of the slotted or elongated openings **2670**, **2672**, **2674**, **2676** extend may vary, having differences in height, length, width, etc. In certain aspects, one or more of the openings **2670**, **2672**, **2674**, **2676** may be omitted. These variations can be realized in any one or more of the other embodiments described herein.

[0263] The openings **2670**, **2672**, **2674**, **2676** are all arranged in an interior area of the resistor **2610**. The openings **2670**, **2672**, **2674**, **2676** are each spaced away from outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **2610**. No portion of the openings **2670**, **2672**, **2674**, **2676** intersect, contact or cross the outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor **2610**. In this way, the openings **2670**, **2672**, **2674**, **2676** are surrounded on all sides by parts of the conductive portions, the resistive elements, or combinations of those.

[0264] As shown in FIGS. 26A-26C, a header assembly **2680** is provided that is configured to provide a convenient arrangement for securing pins, such as a first pin **2684a**, second pin **2684b**, third pin **2684c**, and fourth pin **2684d**. The header assembly **2680** includes at least one brace having a plurality of openings, i.e. a first opening **2681a**, second opening **2681b**, third opening **2681c**, and fourth opening **2681d**, configured to attach to or otherwise secure at least one of the pins **2684a**, **2684b**, **2684c**, **2684d**.

[0265] As shown in FIGS. 26A-26C, the first opening **2681a**, second opening **2681b**, third opening **2681c**, and fourth opening **2681d** are occupied by a respective one of the first pin **2684a**, second pin **2684b**, third pin **2684c**, and fourth pin **2684d**. This configuration is advantageous for manufacturing and allows soldering of the pins **2684a**, **2684b**, **2684c**, **2684d** to inner or inside surfaces of openings **2672**, **2676**, which further improve TCR performance.

[0266] The pins **2684a**, **2684b**, **2684c**, **2684d** are configured to be connected, such as via soldering, to a printed circuit board (PCB) **2695** or a connector that is attached to the PCB. In one aspect, the pins **2684a**, **2684b**, **2684c**, **2684d** are configured as voltage sense pins and are configured to be connected to a positive and negative voltage section of a sensing circuit.

[0267] As shown in FIGS. 24A and 24B, a first header assembly **2480a** and a second header assembly **2480b** are provided that are configured to provide a convenient arrangement for securing pins, such as a first pin **2484a**, second pin **2484b**, third pin **2484c**, fourth pin **2484d**, and fifth pin **2484e**. The header assemblies **2480a**, **2480b** includes braces having a plurality of openings, i.e. a first opening **2481a**, second opening **2481b**, third opening **2481c**, fourth opening **2481d**, fifth opening **2481e**, sixth opening **2481f**, seventh opening **2481g**, eighth opening **2481h**, ninth opening

2481i, tenth opening 2481j, eleventh opening 2481k, and twelfth opening 2481l, configured to attach to or otherwise secure at least one of the pins 2484a, 2484b, 2484c, 2484d, 2484e. As shown in FIG. 24A, only the first opening 2481a, second opening 2481b, sixth opening 2481f, eighth opening 2481h, and twelfth opening 2481l are occupied by a respective one of the pins 2484a, 2484b, 2484c, 2484d, 2484e. This configuration is advantageous for manufacturing and allows soldering of the pins 2484a, 2484b, 2484c, 2484d, 2484e to inner or inside surfaces of openings 2472, 2476, which further improve TCR performance. FIGS. 24A and 24B illustrate a dual current sense resistor, which is considered a redundant voltage connection arrangement in one aspect.

[0268] As shown in FIGS. 24A and 24B, the second, third, fourth, and fifth pins 2484b, 2484c, 2484d, 2484e are configured as sensing pins and the first pin 2484a is configured as a ground pin, which is shown by cross-hatching in FIG. 24B for illustrative purposes only. The sensing pins 2484b, 2484c, 2484d, 2484e are preferably arranged so as to contact inner surface edges of the openings 2472, 2476 that are closest to each other, as shown in FIGS. 24A and 24B. Ground pin 2484a is arranged so as to contact an outermost surface edge of opening 2476. The ground pin 2484a generally provides a ground reference for a measurement system that is separate from the rest of the resistor. The sensing pins 2484b, 2484c, 2484d, 2484e are configured to be connected, such as via soldering, to a printed circuit board (PCB) or connector that is attached to the PCB. In one aspect, the sensing pins 2484b, 2484c, 2484d, 2484e are voltage sense pins and are configured to be connected to a positive and negative voltage section of a sensing circuit.

[0269] Although five pins 2484a, 2484b, 2484c, 2484d, 2484e are illustrated in FIGS. 24A and 24B and the pins 2484a, 2484b, 2484c, 2484d, 2484e are shown in specific configurations, one of ordinary skill in the art would understand that the quantity, location, shape, orientation, and other aspects of the pins 2484a, 2484b, 2484c, 2484d, 2484e can vary.

[0270] FIG. 25 illustrates another resistor 2510 having similar characteristics as the resistor 2110 of FIGS. 21A and 21B. As shown in FIG. 25, the resistor 2510 is generally formed of a resistive element 2511 disposed between a first conductive portion 2512 and a second conductive portion 2514. The resistive element 2511 may be welded or otherwise bonded between the first conductive portion 2512 and the second conductive portion 2514. The resistor 2510 may be similar to the resistor 110 of FIG. 2 and has a similar function and structure to resistor 110 unless otherwise specified.

[0271] As shown in FIG. 25, a first TCR adjustment opening 2572 and a second TCR adjustment opening 2576 are formed on the first conductive portion 2512 and the second conductive portion 2514, and current connections or connection areas are provided via outer circular openings 2570, 2574 on the first conductive portion 2512 and the second conductive portion 2514. As shown in FIG. 25, the first TCR adjustment opening 2572 is formed as an oval, and the second TCR adjustment opening 2576 has an elongated, slot-like profile. FIG. 25 illustrates one aspect of a single current sense resistor, that is, there is a single resistive element.

[0272] All aspects of the openings 2570, 2572, 2574, 2576 may vary. For example, the openings 2570, 2572, 2574, 2576 may each have a different profile, dimension, size, or shape. These may be adjusted or changed based on need or function. The openings 2570, 2572, 2574, 2576 may include one or more through openings and one or more openings that do not extend completely through the conductive portions and/or the resistive elements. The openings 2570, 2572, 2574, 2576 may be circular, elongated, slotted, non-circular, S-shaped, N-shaped, serpentine, or any other shape as may be selected. The orientation or direction in which any of the slotted or elongated openings 2570, 2572, 2574, 2576 extend may vary, having differences in height, length, width, etc. In certain aspects, one or more of the openings 2570, 2572, 2574, 2576 may be omitted. These variations can be realized in any one or more of the other embodiments described herein.

[0273] The openings 2570, 2572, 2574, 2576 are all arranged in an interior area of the resistor 2510. The openings 2570, 2572, 2574, 2576 are each spaced away from outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor 2510. No portion of the openings

2570, 2572, 2574, 2576 intersect, contact or cross the outer edges, such as the lateral and longitudinal edges, or the perimeter of the resistor 2510. In this way, the openings 2570, 2572, 2574, 2576 are surrounded on all sides by parts of the conductive portions, the resistive elements, or combinations of those.

[0274] The resistor shown in FIG. 25 does not include a header or header assembly, as disclosed in FIGS. 21A-24B. Instead, FIG. 25, illustrates a PCB 2595, 2595' that is used in conjunction with the resistor 2510. Instead of using pins that engage an interior of openings formed on the resistor, the configuration of FIG. 25 relies on surface connections between the resistor 2510 and the PCB 2595.

[0275] The PCB is illustrated in a state prior to engaging the resistor 2510 as element 2595' in a bottom region of FIG. 25, and the PCB is illustrated in a state engaging the resistor 2510 as element 2595 in the upper region of FIG. 25.

[0276] Sensing areas 2596a, 2596b, 2596c are provided on the PCB 2595. The sensing areas 2596a, 2596b, 2596c essentially provide the same function as the sensing pins described herein. A connection is provided between the resistor 2510 and the PCB 2595 via engagement of conductor portions 2597a, 2597b formed on the PCB 2595 with the resistor 2510. Areas 2598a, 2598b are non-conductive regions of the PCB 2595. In the embodiment shown in FIG. 25, via on the PCB are used for voltage sensing.

[0277] As shown in the embodiments, the outer openings (i.e. openings 70, 74, 170, 174, 270, 274, 370, 374, 1270, 1274, 1370a-e, 1374a-e, 1770a, 1774b, 1770d, 1774d, 1770e, 1774e, 1770f, 1774f, 1770g, 1774g, 1770h, 1774h, 1770i, 1774i, 1770j, 1774j, 1770k, 1774k, 1774l, 1770m, 1870, 1874, 1970, 1974, 2070, 2074, 2170, 2174, 2270, 2274, 2370, 2374, 2470, 2474, 2570, 2574, 2670, 2674 and the unlabeled outer circular openings in FIGS. 5-8) may function as a current connection hole or area. The current connection openings provide only one example of how a resistor may be attached to a current that is to be monitored and/or measured. Such openings may be omitted entirely.

[0278] As shown in the embodiments, the interior or inner openings (i.e. openings 72, 76, 78, 172, 176, 272, 276, 278, 279, 372, 376, 378, 472, 476, 478, 572, 576, 578, 672, 676, 772, 776, 872, 876, 1272, 1276, 1278, 1372a-1372e, 1376a-1376e, 1378a-1378e, 1772a-1772m, 1776a-1776m, 1778a-1778m, 1872, 1876, 1878, 1972, 1976, 1978, 2072, 2076, 2078, 2172, 2176, 2272, 2276, 2372, 2376, 2472, 2476, 2572, 2576, 2672, 2676) function as TCR adjustment openings, which are also referred to as a TCR compensation slots. The TCR adjustment openings (i.e. openings 72, 76, 78, 172, 176, 272, 276, 278, 279, 372, 376, 472, 476, 572, 576, 578, 672, 676, 772, 776, 872, 876, 1272, 1276, 1278, 1372a-1372e, 1376a-1376e, 1378d, 1772a-1772m, 1776a-1776m, 1778a-1778m, 1872, 1876, 1972, 1976, 2072, 2076, 2172, 2176, 2272, 2276, 2372, 2376, 2472, 2476, 2572, 2576, 2672, 2676) each have a perimeter or perimeter sidewall that is continuously curved, non-linear, and non-circular. In one aspect, the perimeter of the TCR openings is defined by a perimeter wall formed by the resistive elements, conductive portions, and/or combinations thereof, and the perimeter has a portion that is continuously curved, non-linear, and non-circular.

[0279] Certain TCR adjustment openings (i.e. openings 378, 478, 578, 778, 878, 1378a-1378c, 1878, 1978, 2078) are exceptions to the continuously curved, non-linear, and non-circular perimeters for the TCR adjustment openings, and each of these TCR adjustment openings have alternative shaped perimeters or perimeter sidewalls.

[0280] Certain TCR adjustment openings (i.e. openings 272, 276, 278, 279, 372, 376, 472, 476, 572, 576, 672, 676, 772, 776, 872, 876, 1372a-1372c, 1376a-1376c, 1872, 1876, 1972, 1976, 2072, 2076) have a perimeter or perimeter sidewall with a continuous arc profile. In one aspect, the perimeter of the TCR openings is defined by a perimeter wall formed by the resistive elements, conductive portions, and/or combinations thereof, and the perimeter has a portion formed by a continuous arc.

[0281] Certain TCR adjustment openings (i.e. openings 72, 76, 78, 172, 176, 1272, 1276, 1278,

1372d, 1372e, 1376d, 1376e, 1772a-1772m, 1776a-1776m, 1778a-1778m, 2172, 2176, 2272, 2276, 2372, 2376, 2472, 2476, 2572, 2576, 2672, 2676) have a perimeter or perimeter sidewall with an oblong, oval, or oblong oval profile. In one aspect, the perimeter of the TCR openings is defined by a perimeter wall formed by the resistive elements, conductive portions, and/or combinations thereof, and the perimeter has a portion formed with an oblong, oval, or oblong oval profile. [0282] In each of the embodiments described herein, the TCR adjustment openings may be generally located in an interior area of the respective portions, whether in the conductive strips or resistive elements. The TCR adjustment openings are each generally arranged in interior areas of the resistors, in one aspect. The openings in the conductive portions may be surrounded, such as completely surrounded, by the materials of the conductive portions. The opening in the resistive elements may be surrounded, such as completely surrounded, by the materials of the resistive elements or may be surrounded by the materials of the resistive elements and the materials of the conductive portions to the extent the openings extend through parts of the conductive openings. The openings are formed away from an outer perimeter edge of the conductive strips or resistive elements. The openings in the resistive element are formed away from the outer or perimeter edges of the resistive element or resistor.

[0283] Any of the resistors disclosed herein can be used in battery management systems for hybrid-EV, full-EV, hybrid plug-in EV for high, mid-, and low voltage systems, as well as current monitoring in wind generators and other alternative energy generation. The resistors could also be used in direct current sensing, current sensing in conjunction to and attached to other circuitry, as well as the automotive, industrial, and renewable energy industries.

[0284] Although the features and elements of the present invention are described in the example embodiments in particular combinations, each feature may be used alone without the other features and elements of the example embodiments or in various combinations with or without other features and elements of the present invention.

[0285] In the drawings and specification there has been set forth embodiments of the invention, and although specific terms are employed these are used in a generic and descriptive sense only and not for purposes of limitation. Changes in the form and the proportion of parts as well as in the substitution of equivalents are contemplated as circumstances may suggest or render expedient without departing from the spirit or scope of the invention as further defined in the following claims.

[0286] The foregoing descriptions of specific embodiments of the present technology have been presented for purposes of illustration and description. They are not intended to be exhaustive or to limit the invention to the precise forms disclosed, and obviously many modifications and variations are possible in light of the above teaching. The embodiments were chosen and described in order to best explain the principles of the present technology and its practical application, to thereby enable others skilled in the art to best utilize the present technology and various embodiments with various modifications as are suited to the particular use contemplated. It is intended that the scope of the invention be defined by the claims appended hereto and their equivalents.

[0287] The present disclosure is not limited to the precise construction and compositions disclosed herein. Moreover, the present concepts expressly include any and all combinations and subcombinations of the preceding elements and features.

Claims

1. A resistor comprising: a first resistive element disposed between a first conductive portion and a second conductive portion; the first conductive portion having a first side and a second side, the second side being positioned closer to the first resistive element than the first side; the second conductive portion having a first side and a second side, the second side being positioned closer to the first resistive element than the first side; a first temperature coefficient of resistance (TCR)

adjustment opening positioned adjacent to the second side of the first conductive portion in a first interior area of the resistor, the first TCR adjustment opening formed with a first profile having a continuously curved, non-linear, and non-circular perimeter; and a second temperature coefficient of resistance (TCR) adjustment opening positioned adjacent to the second side of the second conductive portion in a second interior area of the resistor, the second TCR adjustment opening formed with a second profile having a continuously curved, non-linear, and non-circular perimeter; wherein the first profile of the first TCR adjustment opening and the second profile of the second TCR adjustment opening are each configured to adjust a TCR value of the resistor.

2. The resistor according to claim 1, wherein the first TCR adjustment opening and the second TCR adjustment opening are each formed as an elongated slot with a continuous arc profile.
3. The resistor according to claim 1, wherein the first TCR adjustment opening and the second TCR adjustment opening each have an oblong or oval profile.
4. The resistor according to claim 1, wherein terminal ends of the first TCR adjustment opening are curved towards a same longitudinal direction as each other, and terminal ends of the second TCR adjustment opening are curved towards a same longitudinal direction as each other.
5. The resistor according to claim 4, wherein the first TCR adjustment opening and the second TCR adjustment opening each have an arc length of greater than 90 degrees.
6. The resistor according to claim 1, wherein the first TCR adjustment opening and the second TCR adjustment opening are symmetrical and mirror images of each other.
7. The resistor according to claim 1, wherein the first TCR adjustment opening and the second TCR adjustment opening each have concave profiles relative to a direction of the first resistive element.
8. The resistor according to claim 1, further comprising a third TCR adjustment opening, the third TCR adjustment opening being partially defined on each of the first resistive element, the first conductive portion, and the second conductive portion, wherein the third TCR adjustment opening is configured to adjust the TCR value of the resistor.
9. The resistor according to claim 8, wherein the third TCR adjustment opening has an oblong profile, or a profile having a smaller central area and larger end areas.
10. The resistor according to claim 8, wherein the third TCR adjustment opening extends for an entire longitudinal extent of the first resistive element.
11. The resistor according to claim 8, wherein a first end of the third TCR adjustment opening is spaced apart in a longitudinal direction from the first TCR adjustment opening, and a second end of the third TCR adjustment opening is spaced apart in the longitudinal direction from the second TCR adjustment opening.
12. The resistor according to claim 8, wherein a first end of the third TCR adjustment opening is partially surrounded by the first TCR adjustment opening, and a second end of the third TCR adjustment opening is partially surrounded by the second TCR adjustment opening.
13. A resistor comprising: a first resistive element disposed between a first conductive portion and a second conductive portion; a second resistive element disposed between the second conductive portion and a third conductive portion; a first temperature coefficient of resistance (TCR) adjustment opening arranged on the first conductive portion and positioned adjacent to the first resistive element in a first interior area of the resistor; a second temperature coefficient of resistance (TCR) adjustment opening arranged on the third conductive portion and positioned adjacent to the second resistive element in a second interior area of the resistor; a third temperature coefficient of resistance (TCR) adjustment opening arranged on the second conductive portion and positioned between the first resistive element and the second resistive element in a third interior area of the resistor; wherein profiles of the first TCR adjustment opening, the second TCR adjustment opening, and the third TCR adjustment opening are each configured to adjust a TCR value of the resistor.
14. The resistor according to claim 13, wherein the first TCR adjustment opening and the second TCR adjustment opening each have a C-shaped profile.

- 15.** The resistor according to claim 13, wherein the first TCR adjustment opening and the second TCR adjustment opening each have an oblong or oval profile.
- 16.** The resistor according to claim 13, wherein the third TCR adjustment opening has a profile having a smaller central area and larger end areas.
- 17.** The resistor according to claim 13, wherein the first TCR adjustment opening and the second TCR adjustment opening have an identical profile, and the third TCR adjustment opening has a different profile from the profiles of the first TCR adjustment opening and the second TCR adjustment opening.
- 18.** The resistor according to claim 13, wherein terminal ends of the first TCR adjustment opening are curved towards a same longitudinal direction as each other, and terminal ends of the second TCR adjustment opening are curved towards a same longitudinal direction as each other.
- 19.** A method of forming a resistor, the method comprising: providing a first resistive element disposed between a first conductive portion and a second conductive portion; forming a first temperature coefficient of resistance (TCR) opening in the first conductive portion in a first interior area of the resistor; and forming a second temperature coefficient of resistance (TCR) opening in the second conductive portion in a second interior area of the resistor; wherein the first TCR adjustment opening and the second TCR adjustment opening are each formed with a continuously curved, non-linear, and non-circular perimeter.
- 20.** The method according to claim 19, further comprising adjusting at least one characteristic of at least one of the first TCR opening or the second TCR opening to provide for an adjusted TCR value of the resistor.
- 21.** (canceled)
- 22.** (canceled)
- 23.** (canceled)
- 24.** (canceled)
- 25.** (canceled)
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